

BOOK 1

PHYSICAL

QUANTITIES & UNITS

1 Physical quantities and units

1.1 Physical quantities

Candidates should be able to:

- 1 understand that all physical quantities consist of a numerical magnitude and a unit
- 2 make reasonable estimates of physical quantities included within the syllabus

1.2 SI units

Candidates should be able to:

- 1 recall the following SI base quantities and their units: mass (kg), length (m), time (s), current (A), temperature (K)
- 2 express derived units as products or quotients of the SI base units and use the derived units for quantities listed in this syllabus as appropriate
- 3 use SI base units to check the homogeneity of physical equations
- 4 recall and use the following prefixes and their symbols to indicate decimal submultiples or multiples of both base and derived units: pico (p), nano (n), micro (μ), milli (m), centi (c), deci (d), kilo (k), mega (M), giga (G), tera (T)

1.4 Scalars and vectors

Candidates should be able to:

- 1 understand the difference between scalar and vector quantities and give examples of scalar and vector quantities included in the syllabus
- 2 add and subtract coplanar vectors
- 3 represent a vector as two perpendicular components

1 9702/1/M/J/02/Q1

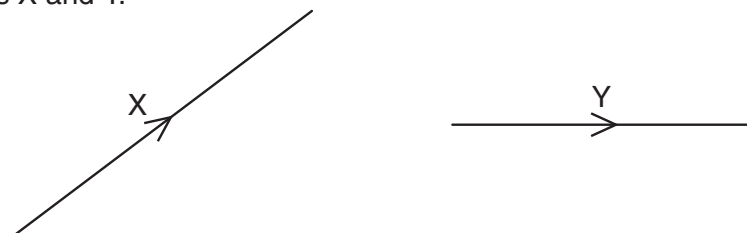
Which of the following pairs of units are both SI base units?

A ampere, degree celsius **C** coulomb, degree celsius

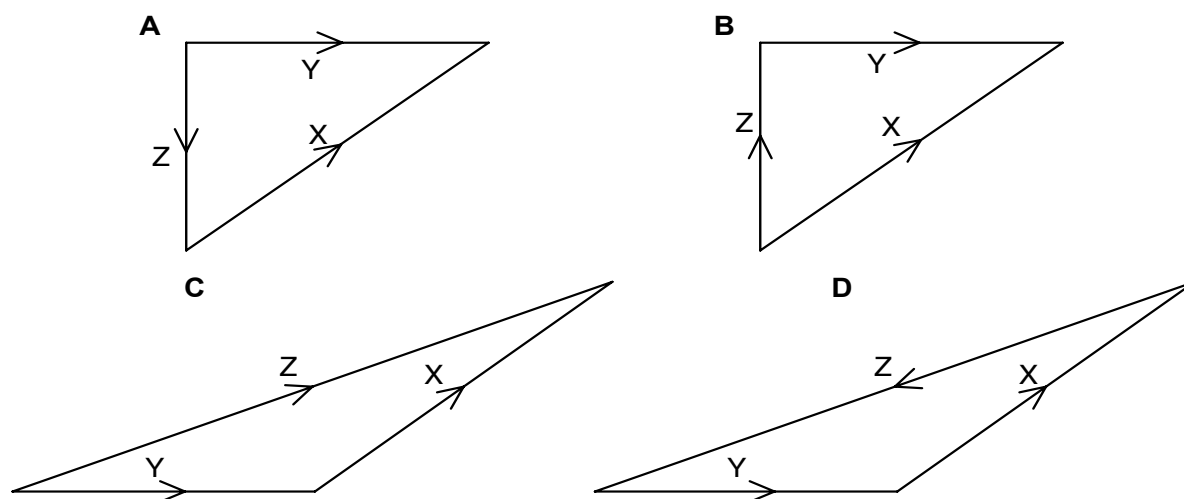
B ampere, kelvin **D** coulomb, kelvin

2 9702/1/M/J/02/Q2*

The diagram shows two vectors X and Y.



In which vector triangle does the vector Z show the magnitude and direction of vector $X - Y$?



3 9702/1/O/N/02/Q1

The prefix 'centi' indicates $\times 10^{-2}$. That is, 1 centimetre is equal to 1×10^{-2} metre.

Which line in the table correctly indicates the prefixes micro, nano and pico?

	$\times 10^{-12}$	$\times 10^{-9}$	$\times 10^{-6}$
A	nano	micro	pico
B	micro	pico	nano
C	pico	nano	micro
D	pico	micro	nano

4 9702/01/M/J/03/Q1

Which of the following is a scalar quantity?

A acceleration **B** mass **C** momentum **D** velocity

5 9702/01/M/J/03/Q2

The unit of work, the joule, may be defined as the work done when the point of application of a force of 1 newton is moved a distance of 1 metre in the direction of the force.

Express the joule in terms of the base units of mass, length and time, the kg, m and s.

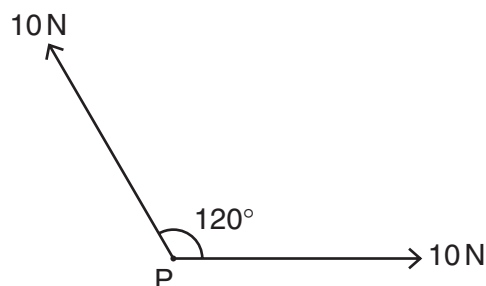
- A $\text{kg m}^{-1} \text{s}^2$ B $\text{kg m}^2 \text{s}^{-2}$ C $\text{kg m}^2 \text{s}^{-1}$ D kg s^{-2}

6 9702/01/M/J/03/Q3

Two forces, each of 10 N, act at a point P as shown in the diagram. The angle between the directions of the forces is 120° .

What is the magnitude of the resultant force?

- A 5 N B 10 N C 17 N D 20 N



7 9702/01/O/N/03/Q1

A student measures a current as 0.5 A.

Which of the following correctly expresses this result?

- A 50 mA B 50 MA C 500 mA D 500 MA

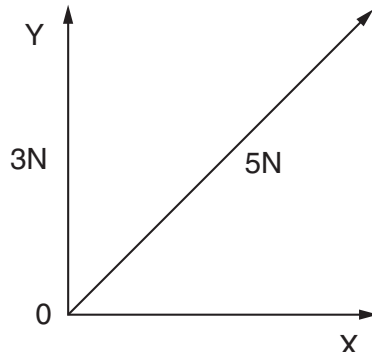
8 9702/01/O/N/03/Q2

A force of 5 N may be represented by two perpendicular components OY and OX as shown in the diagram, which is **not** drawn to scale.

OY is of magnitude 3 N.

What is the magnitude of OX?

- A 2 N B 3 N C 4 N D 5 N



9 9702/01/O/N/03/Q3

The momentum of an object of mass m is p .

Which quantity has the same base units as $\frac{p^2}{m}$?

- A energy B force C power D velocity

10 9702/11/M/J/15/Q2*

Which pair contains one vector and one scalar quantity?

- | | | | | | |
|----------------|---|----------------|------------|---|----------|
| A displacement | : | acceleration | C momentum | : | velocity |
| B force | : | kinetic energy | D power | : | speed |

11 9702/01/M/J/04/Q2

Which of the following could be measured in the same units as force?

- | | |
|---------------------|-----------------------|
| A energy / distance | C energy / time |
| B energy x distance | D momentum x distance |

12 9702/01/M/J/04/Q3

The notation μs is used as an abbreviation for a certain unit of time.

What is the name and value of this unit?

	name	value
A	microsecond	10^{-6}s
B	microsecond	10^{-3}s
C	millisecond	10^{-6}s
D	millisecond	10^{-3}s

13 9702/01/O/N/04/Q1

Which line of the table gives values that are equal to a time of 1 ps (one picosecond) and a distance of 1 Gm (one gigametre)?

	time of 1 ps	distance of 1 Gm
A	10^{-9}s	10^9m
B	10^{-9}s	10^{12}m
C	10^{-12}s	10^9m
D	10^{-12}s	10^{12}m

14 9702/01/O/N/04/Q2

Which of the following definitions is correct and uses only quantities rather than units?

- A** Density is mass per cubic metre.
- B** Potential difference is energy per unit current.
- C** Pressure is force per unit area.
- D** Speed is distance travelled per second.

15 9702/01/O/N/04/Q3

When a beam of light is incident on a surface, it delivers energy to the surface. The intensity of the beam is defined as the energy delivered per unit area per unit time.

What is the unit of intensity, expressed in SI base units?

- A** $\text{kg m}^{-2} \text{s}^{-1}$
- B** $\text{kg m}^2 \text{s}^{-3}$
- C** kg s^{-2}
- D** kg s^{-3}

16 9702/01/M/J/05/Q1*

Decimal sub-multiples and multiples of units are indicated using a prefix to the unit. For example, the prefix milli (m) represents 10^{-3} .

Which of the following gives the sub-multiples or multiples represented by pico (p) and giga (G)?

	pico (p)	giga (G)
A	10^{-9}	10^9
B	10^{-9}	10^{12}
C	10^{-12}	10^9
D	10^{-12}	10^{12}

17 9702/01/M/J/05/Q2*

A metal sphere of radius r is dropped into a tank of water. As it sinks at speed v , it experiences a drag force F given by $F = krv$, where k is a constant.

What are the SI base units of k ?

- A** $\text{kg m}^2 \text{s}^{-1}$ **B** $\text{kg m}^{-2} \text{s}^{-2}$ **C** $\text{kg m}^{-1} \text{s}^{-1}$ **D** kg ms^{-2}

18 9702/11/O/N/10/Q2*

An Olympic athlete of mass 80 kg competes in a 100 m race.

What is the best estimate of his mean kinetic energy during the race?

- A** $4 \times 10^2 \text{ J}$ **B** $4 \times 10^3 \text{ J}$ **C** $4 \times 10^4 \text{ J}$ **D** $4 \times 10^5 \text{ J}$

19 9702/01/O/N/05/Q1

Which pair of units are both SI base units?

- A** ampere, degree celsius **C** coulomb, degree celsius
B ampere, kelvin **D** coulomb, kelvin

20 9702/01/O/N/05/Q2

The prefix 'centi' indicates $\times 10^{-2}$.

Which line in the table correctly indicates the prefixes micro, nano and pico?

	$\times 10^{-12}$	$\times 10^{-9}$	$\times 10^{-6}$
A	nano	micro	pico
B	nano	pico	micro
C	pico	nano	micro
D	pico	micro	nano

21 9702/01/O/N/05/Q3

Which expression involving base units is equivalent to the volt?

- A** $\text{kg m}^2 \text{s}^{-1} \text{A}^{-1}$ **B** $\text{kg ms}^{-2} \text{A}$ **C** $\text{kg m}^2 \text{s}^{-1} \text{A}$ **D** $\text{kg m}^2 \text{s}^{-3} \text{A}^{-1}$

22 9702/12/M/J/13/Q1**

Which pair includes a vector quantity and a scalar quantity?

- A** displacement; acceleration **C** power; speed
B force; kinetic energy **D** work; potential energy

23 9702/12/M/J/12/Q2

For which quantity is the magnitude a reasonable estimate?

- A** frequency of a radio wave 500 pHz **C** the Young modulus of a metal 500 kPa
B mass of an atom 500 μg **D** wavelength of green light 500 nm

24 9702/01/M/J/06/Q3*

The following physical quantities can be either positive or negative.

s : displacement of a particle along a straight line

q : electric charge

θ : temperature on the Celsius scale

V : readings on a digital voltmeter

Which of these quantities are vectors?

- A** s, θ, q, V **B** s, q, V **C** θ, V **D** s only

25 9702/01/O/N/06/Q1

Which product-pair of metric prefixes has the greatest magnitude?

- A** pico × mega **B** nano × kilo **C** micro × giga **D** milli × tera

26 9702/01/O/N/06/Q2

In the expressions below

a is acceleration, F is force, m is mass, t is time, v is velocity.
Which expression represents energy?

- A** Ft **B** Fvt **C** $\frac{2mv}{t}$ **D** $\frac{at^2}{2}$

27 9702/01/O/N/06/Q3

Which row of the table shows a physical quantity and its correct unit?

	physical quantity	unit
A	electric field strength	$\text{kg m s}^{-2} \text{C}^{-1}$
B	specific heat capacity	$\text{kg}^{-1} \text{m}^2 \text{s}^{-2} \text{K}^{-1}$
C	tensile strain	$\text{kg m}^{-1} \text{s}^{-2}$
D	the Young modulus	$\text{kg m}^{-1} \text{s}^{-3}$

28 9702/01/M/J/07/Q1

Which is a pair of SI base units?

A	ampere	joule
B	coulomb	second
C	kilogram	kelvin
D	metre	newton

29 9702/01/M/J/07/Q2

What is the ratio $\frac{1\mu\text{m}}{1\text{Gm}}$?

- A** 10^{-3} **B** 10^{-9} **C** 10^{-12} **D** 10^{-15}

30 9702/01/M/J/07/Q3

Which formula could be correct for the speed v of ocean waves in terms of the density ρ of sea-water, the acceleration of free fall g , the depth h of the ocean and the wavelength λ ?

- A** $v = \sqrt{g\lambda}$ **B** $v = \sqrt{\frac{g}{h}}$ **C** $v = \sqrt{\rho gh}$ **D** $v = \sqrt{\frac{g}{\rho}}$

31 9702/01/O/N/07/Q1

The equation relating pressure and density is $p = \rho gh$.

How can both sides of this equation be written in terms of base units?

- A** $[\text{N m}^{-1}] = [\text{kg m}^{-3}] [\text{m s}^{-1}] [\text{m}]$ **C** $[\text{kg m}^{-1} \text{s}^{-2}] = [\text{kg m}^{-3}] [\text{m s}^{-2}] [\text{m}]$
B $[\text{N m}^{-2}] = [\text{kg m}^{-3}] [\text{m s}^{-2}] [\text{m}]$ **D** $[\text{kg m}^{-1} \text{s}^{-1}] = [\text{kg m}^{-1}] [\text{m s}^{-2}] [\text{m}]$

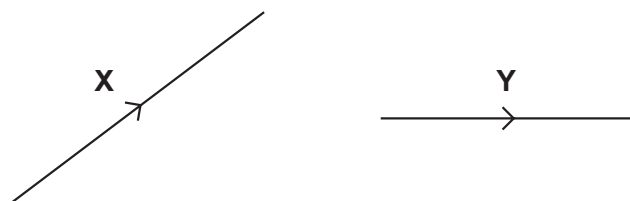
32 9702/01/O/N/07/Q2

What is a reasonable estimate of the diameter of an alpha particle?

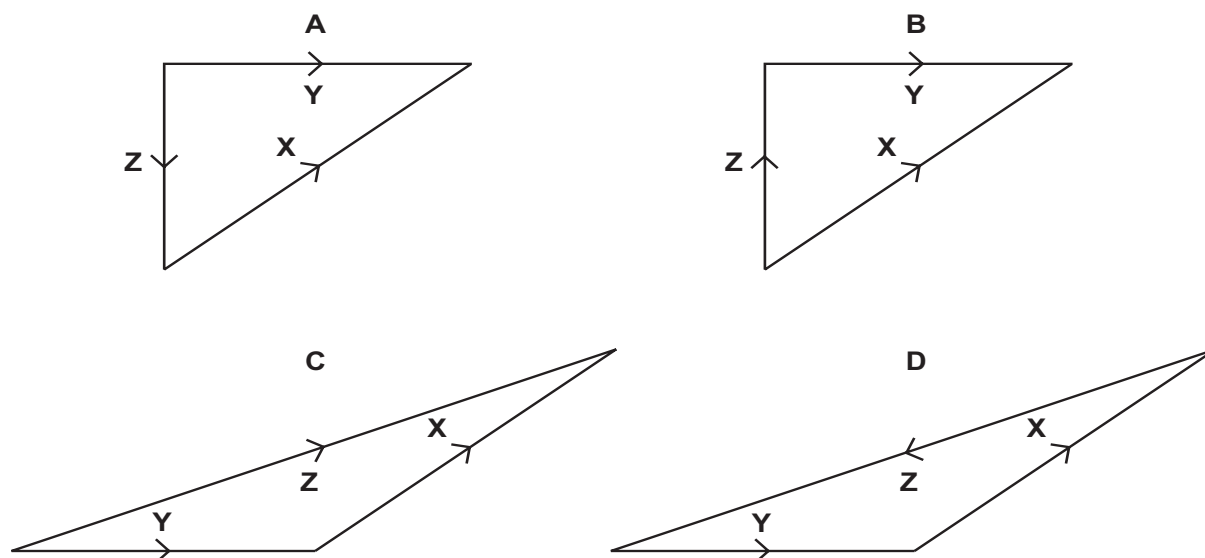
- A** 10^{-15}m **B** 10^{-12}m **C** 10^{-9}m **D** 10^{-6}m

33 9702/01/O/N/07/Q3

The diagram shows two vectors **X** and **Y**.



In which vector triangle does the vector **Z** show the magnitude and direction of vector **X**−**Y**?



34 9702/01/M/J/08/Q1

Five energies are listed.

5 kJ

5 mJ

5 MJ

5 nJ

Starting with the smallest first, what is the order of increasing magnitude of these energies?

A 5 kJ → 5 mJ → 5 MJ → 5 nJ

C 5 nJ → 5 mJ → 5 kJ → 5 MJ

B 5 nJ → 5 kJ → 5 MJ → 5 mJ

D 5 mJ → 5 nJ → 5 kJ → 5 MJ

35 9702/01/M/J/08/Q2

Which of the following correctly expresses the volt in terms of SI base units?

A $A\Omega$

B WA^{-1}

C $kg\,m^2\,s^{-1}\,A^{-1}$

D $kg\,m^2\,s^{-3}\,A^{-1}$

36 9702/13/M/J/12/Q1*

What is a reasonable estimate of the average kinetic energy of an athlete during a 100 m race that takes 10 s?

A 40 J

B 400 J

C 4000 J

D 40 000 J

37 9702/01/O/N/08/Q1

A laser emits light of wavelength 600 nm.

What is the distance, expressed as a number of wavelengths, travelled by the light in one second?

A 5×10^8

B 5×10^{11}

C 5×10^{14}

D 5×10^{17}

38 9702/01/O/N/08/Q2

At temperatures close to 0 K, the specific heat capacity c of a particular solid is given by $c = bT^3$, where T is the thermodynamic temperature and b is a constant characteristic of the solid.

What are the units of constant b , expressed in SI base units?

- A $\text{m}^2\text{s}^{-2}\text{K}^{-3}$ B $\text{m}^2\text{s}^{-2}\text{K}^{-4}$ C $\text{kg m}^2\text{s}^{-2}\text{K}^{-3}$ D $\text{kg m}^2\text{s}^{-2}\text{K}^{-4}$

39 9702/01/O/N/08/Q3

The table shows the x-component and y-component of four force vectors.

Which force vector has the largest magnitude?

	x-component / N	y-component / N
A	2	9
B	3	8
C	4	7
D	5	6

40 9702/01/M/J/09/Q1

Which statement, involving multiples and sub-multiples of the base unit metre (m), is correct?

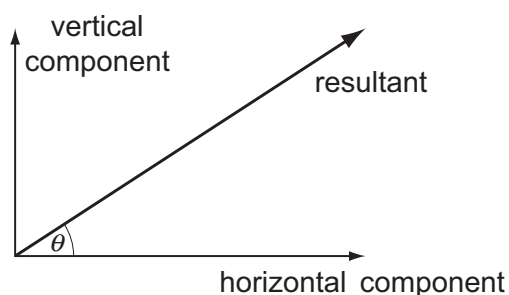
- A $1\text{ pm} = 10^{-9}\text{ m}$ B $1\text{ nm} = 10^{-6}\text{ m}$ C $1\text{ mm} = 10^6\text{ }\mu\text{m}$ D $1\text{ km} = 10^6\text{ mm}$

41 9702/01/M/J/09/Q2

The diagram shows a resultant force and its horizontal and vertical components.

The horizontal component is 20.0 N and $\theta = 30^\circ$

What is the vertical component?



- A 8.7 N B 10.0 N
C 11.5 N D 17.3 N

42 9702/12/O/N/09/Q1

The drag force F acting on a moving sphere obeys an equation of the form $F = kAv^2$, where A represents the sphere's frontal area and v represents its speed.

What are the base units of the constant k ?

- A $\text{kg m}^5\text{s}^{-4}$ B $\text{kg m}^{-2}\text{s}^{-1}$ C kg m^{-3} D $\text{kg m}^{-4}\text{s}^2$

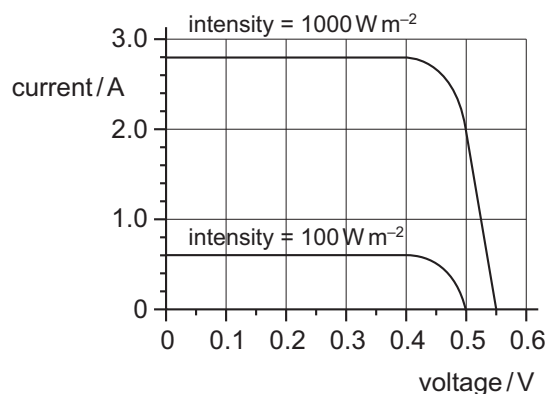
43 9702/12/O/N/09/Q2

The graph shows two current-voltage calibration curves for a solar cell exposed to different light intensities.

At zero voltage, what is the ratio

$$\frac{\text{current at } 1000\text{ W m}^{-2}}{\text{current at } 100\text{ W m}^{-2}}?$$

- A 1.1 B 4.7
C 8.0 D 10



44 9702/12/O/N/09/Q40

The table contains some quantities, together with their symbols and units.

Which expression has the units of energy?

- A $g\rho hV$ B $\frac{\rho hV}{g}$
 C $\frac{\rho g}{hV}$ D $\rho g^2 h$

quantity	symbol	unit
gravitational field strength	g	N kg^{-1}
density of liquid	ρ	kg m^{-3}
vertical height	h	m
volume of part of liquid	V	m^3

45 9702/12/M/J/10/Q1

A micrometer screw gauge is used to measure the diameter of a copper wire.

The reading with the wire in position is shown in diagram 1. The wire is removed and the jaws of the micrometer are closed. The new reading is shown in diagram 2.

What is the diameter of the wire?

- A 1.90 mm B 2.45 mm
 C 2.59 mm D 2.73 mm

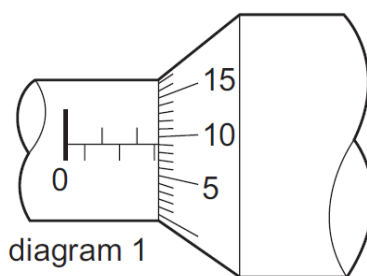


diagram 1

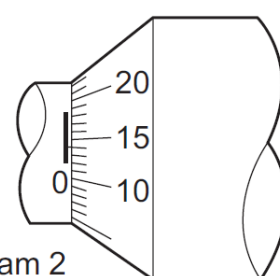


diagram 2

46 9702/12/M/J/10/Q2

The SI unit for potential difference (the volt) is given, in base units, by

- A $\text{kg m A}^{-1} \text{s}^{-3}$ B $\text{m}^2 \text{A}^{-1} \text{s}^{-2}$ C $\text{kg m}^2 \text{s}^{-2}$ D $\text{kg m}^2 \text{A}^{-1} \text{s}^{-3}$

47 9702/12/M/J/10/Q7

The product of pressure and volume has the same SI base units as

- A energy. B force. C $\frac{\text{force}}{\text{area}}$ D $\frac{\text{force}}{\text{length}}$

48 9702/11/O/N/10/Q1

A signal has a frequency of 2.0 MHz.

What is the period of the signal?

- A $2 \mu\text{s}$ B $5 \mu\text{s}$ C 200 ns D 500 ns

49 9702/01/M/J/05/Q2

A metal sphere of radius r is dropped into a tank of water. As it sinks at speed v , it experiences a drag force F given by $F = krv$, where k is a constant.

What are the SI base units of k ?

- A $\text{kg m}^2 \text{s}^{-1}$ B $\text{kg m}^{-2} \text{s}^{-2}$ C $\text{kg m}^{-1} \text{s}^{-1}$ D kg m s^{-2}

50 9702/11/O/N/10/Q3

Which physical quantity would result from a calculation in which a potential difference is multiplied by an electric charge?

- A electric current B electric energy C electric field strength D electric power

51 9702/12/O/N/10/Q1

Which row shows a base quantity with its correct SI unit?

	quantity	unit
A	current	A
B	mass	g
C	temperature	$^{\circ}\text{C}$
D	weight	N

52 9702/12/O/N/10/Q2

The frictional force F on a sphere falling through a fluid is given by the formula $F = 6\pi a\eta v$ where a is the radius of the sphere, η is a constant relating to the fluid and v is the velocity of the sphere.

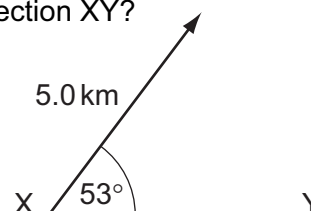
What are the units of η ?

- A kg m s^{-1} B $\text{kg m}^{-1} \text{s}^{-1}$ C kg m s^{-3} D $\text{kg m}^3 \text{s}^{-3}$

53 9702/12/O/N/10/Q3

What is the component of this displacement vector in the direction XY?

- A 3.0 km B 4.0 km
C 5.0 km D 6.6 km



54 9702/11/M/J/11/Q1

Decimal sub-multiples and multiples of units are indicated using a prefix to the unit. For example, the prefix milli (m) represents 10^{-3} .

Which row gives the sub-multiples or multiples represented by pico (p) and giga (G)?

	pico (p)	giga (G)
A	10^{-9}	10^9
B	10^{-9}	10^{12}
C	10^{-12}	10^9
D	10^{-12}	10^{12}

55 9702/11/M/J/11/Q2*

Which definition is correct and uses only quantities rather than units?

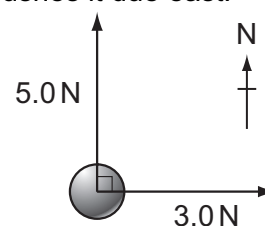
- A Density is mass per cubic metre.
B Potential difference is energy per unit current.
C Pressure is force per unit area.
D Speed is distance travelled per second.

56 9702/11/M/J/11/Q3

A force of 5.0 N pushes a ball due north and another force of 3.0 N pushes it due east.

What is the magnitude of the net force acting on the ball?

- A 2.8 N B 4.0 N
C 5.8 N D 8.0 N



57 9702/12/M/J/11/Q1

Stress has the same SI base units as

- A $\frac{\text{force}}{\text{mass}}$ B $\frac{\text{force}}{\text{length}}$ C $\frac{\text{force}}{\text{area}}$ D energy.

58 9702/12/M/J/11/Q2

To check calculations, the units are put into the following equations together with the numbers. Which equation must be **incorrect**?

- A force = $300 \text{ J} / 6 \text{ m}$ C time = $6 \text{ m} / 30 \text{ m s}^{-1}$
B power = $6000 \text{ J} \times 20 \text{ s}$ D velocity = $4 \text{ m s}^{-2} \times 30 \text{ s}$

59 9702/12/M/J/11/Q3

In making reasonable estimates of physical quantities, which statement is **not** correct?

- A The frequency of sound can be of the order of GHz.
- B The wavelength of light can be of the order of 600 nm.
- C The Young modulus can be of the order of 10^{11} Pa.
- D Beta radiation is associated with one unit of negative charge.

60 9702/11/O/N/11/Q1

Which statement using prefixes of the base unit metre (m) is **not** correct?

- A $1 \text{ pm} = 10^{-12} \text{ m}$
- B $1 \text{ nm} = 10^{-9} \text{ m}$
- C $1 \text{ Mm} = 10^6 \text{ m}$
- D $1 \text{ Gm} = 10^{12} \text{ m}$

61 9702/01/M/J/05/Q3

An Olympic athlete of mass 80 kg competes in a 100 m race.

What is the best estimate of his mean kinetic energy during the race?

- A $4 \times 10^2 \text{ J}$
- B $4 \times 10^3 \text{ J}$
- C $4 \times 10^4 \text{ J}$
- D $4 \times 10^5 \text{ J}$

62 9702/11/O/N/11/Q3

Which group of quantities contains only vectors?

- A acceleration, displacement, speed
- B acceleration, work, electric field strength
- C displacement, force, velocity
- D power, electric field strength, force

63 9702/13/M/J/14/Q1*

Which quantity can be measured in electronvolts (eV)?

- A electric charge
- B electric potential
- C energy
- D power

64 9702/12/O/N/11/Q3

The following physical quantities can be either positive or negative.

s : displacement of a particle along a straight line

θ : temperature on the Celsius scale

q : electric charge

V : readings on a digital voltmeter

Which of these quantities are vectors?

- A s, θ, q, V
- B s, q, V only
- C θ, V only
- D s only

65 9702/12/M/J/12/Q1

What is the unit watt in terms of SI base units?

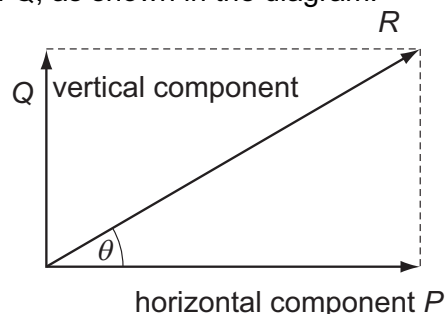
- A J s^{-1}
- B $\text{m}^2 \text{kg s}^{-1}$
- C $\text{m}^2 \text{kg s}^{-3}$
- D N m s^{-1}

66 9702/12/M/J/12/Q3

A vector has magnitude R and perpendicular components P and Q , as shown in the diagram.

Which row correctly describes the perpendicular components?

	vertical component	horizontal component
A	Q	$R \sin \theta$
B	$R \cos \theta$	P
C	$R \cos \theta$	$R \sin \theta$
D	$R \sin \theta$	$R \cos \theta$



67 9702/13/M/J/12/Q1

What is a reasonable estimate of the average kinetic energy of an athlete during a 100 m race that takes 10 s?

- A 40 J
- B 400 J
- C 4000 J
- D 40 000 J

68 9702/13/O/N/15/Q3

When a force F moves its point of application through a displacement s in the direction of the force, the work W done by the force is given by $W = Fs$.

How many vector quantities and scalar quantities does this equation contain?

- A one scalar quantity and two vector quantities C three scalar quantities
B one vector quantity and two scalar quantities D three vector quantities

69 9702/13/M/J/12/Q3

What is a possible unit for the product VI , where V is the potential difference across a resistor and I is the current through the same resistor?

- A newton per second (N s^{-1}) C newton metre (N m)
B newton second (N s) D newton metre per second (N m s^{-1})

70 9702/12/O/N/12/Q1

Which quantity has the same base units as momentum?

- A density $\times$ energy B density $\times$ volume $\times$ velocity C pressure $\times$ area D weight $\div$ area

71 9702/12/O/N/12/Q2

Vectors P and Q are drawn to scale.



Which diagram represents the vector $(P + Q)$?



72 9702/12/O/N/12/Q3

What is the approximate kinetic energy of an Olympic athlete when running at maximum speed during a 100 m race?

- A 400 J B 4000 J C 40 000 J D 400 000 J

73 9702/12/O/N/12/Q4

Physical quantities can be classed as vectors or as scalars.

Which pair of quantities are both vectors?

- A kinetic energy and elastic force C velocity and electric field strength
B momentum and time D weight and temperature

74 9702/13/O/N/12/Q1

The units of all physical quantities can be expressed in terms of SI base units.

Which pair contains quantities with the same base units?

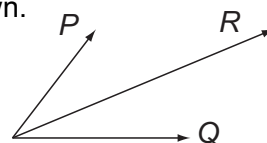
- A force and momentum C power and kinetic energy
B pressure and Young modulus D mass and weight

75 9702/13/O/N/12/Q2

Two physical quantities P and Q are added. The sum of P and Q is R , as shown.

Which quantity could be represented by P and by Q ?

- A kinetic energy B power C speed D velocity



76 9702/13/O/N/12/Q4

Three of these quantities have the same unit.

Which quantity has a different unit?

- A $\frac{\text{energy}}{\text{distance}}$ B force C power $\times$ time D rate of change of momentum

77 9702/11/M/J/13/Q1

Which pair of quantities contains one vector and one scalar quantity?

- A displacement; force C acceleration; momentum
B kinetic energy; power D velocity; distance

78 9702/11/M/J/13/Q2

One property Q of a material is used to describe the behaviour of sound waves in the material. Q is defined as the pressure P of the sound wave divided by the speed v of the wave and the surface area A of the material through which the wave travels:

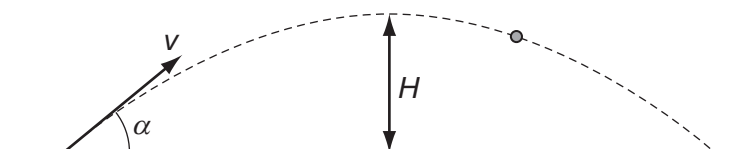
$$Q = \frac{P}{vA}.$$

What are the SI base units of Q ?

- A $\text{kg m}^2 \text{s}^{-3}$ B $\text{kg m}^{-3} \text{s}^{-1}$ C $\text{kg m}^{-4} \text{s}^{-1}$ D $\text{kg m}^{-2} \text{s}^{-2}$

79 9702/11/M/J/13/Q3

A cannon fires a cannonball with an initial speed v at an angle α to the horizontal.



Which equation is correct for the maximum height H reached?

- A $H = \frac{v \sin \alpha}{2g}$ B $H = \frac{g \sin \alpha}{2v}$ C $H = \frac{(v \sin \alpha)^2}{2g}$ D $H = \frac{g^2 \sin \alpha}{2v}$

80 9702/11/M/J/13/Q4

A wave has a frequency of 5 GHz.

What is the period of the wave?

- A 20 000 μs B 20 ns C 2 ns D 200 ps

81 9702/12/M/J/13/Q1

Which pair includes a vector quantity and a scalar quantity?

- A displacement; acceleration C power; speed
B force; kinetic energy D work; potential energy

82 9702/12/M/J/13/Q2

The unit of resistivity, expressed in terms of base units, is given by

Which base units are x , y and z ? $\text{kg x}^3 \text{y}^{-2} \text{z}^{-3}$.

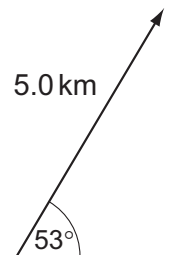
	x	y	z
A	ampere	metre	second
B	metre	ampere	second
C	metre	second	ampere
D	second	ampere	metre

83 9702/13/M/J/13/Q1

The diagram shows a displacement vector.

What is the vertical component of this displacement vector?

- A 3.0 km B 4.0 km C 5.0 km D 6.6 km



84 9702/13/M/J/13/Q2

What is the unit of power, expressed in SI base units?

- A $\text{kg m}^2 \text{s}^{-3}$ B kg m s^{-3} C kg m s^{-2} D $\text{kg m}^2 \text{s}^{-1}$

85 9702/12/O/N/13/Q1

Which row shows an SI base quantity with its correct unit?

	SI base quantity	unit
A	charge	coulomb
B	current	ampere
C	potential difference	volt
D	temperature	degree Celsius

86 9702/12/O/N/13/Q2

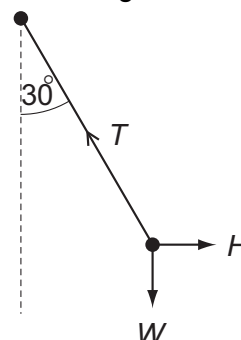
A pendulum bob is held stationary by a horizontal force H . The three forces acting on the bob are shown in the diagram.

The tension in the string of the pendulum is T

The weight of the pendulum bob is W .

Which statement is correct?

- A $H = T \cos 30^\circ$ C $W = T \cos 30^\circ$
 B $T = H \sin 30^\circ$ D $W = T \sin 30^\circ$



87 9702/12/O/N/13/Q3*

The drag coefficient C_d is a number with no units. It is used to compare the drag on different cars at different speeds. It is given by the equation

$$C_d = \frac{2F}{\rho v^n A}$$

where F is the drag force on the car, ρ is the density of the air, A is the cross-sectional area of the car and v is the speed of the car.

What is the value of n ?

- A 1 B 2 C 3 D 4

88 9702/13/O/N/13/Q1

Which estimate is realistic?

- A The kinetic energy of a bus travelling on an expressway is 30 000 J.
 B The power of a domestic light is 300 W.
 C The temperature of a hot oven is 300 K.
 D The volume of air in a car tyre is 0.03 m^3 .

89 9702/13/O/N/13/Q2*

Which unit is equivalent to the coulomb?

- A ampere per second C watt per ampere
 B joule per volt D watt per volt

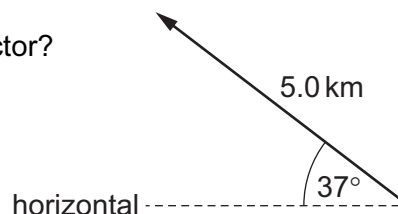
- 90 9702/13/O/N/13/Q3***
Two forces of equal magnitude are represented by two coplanar vectors. One is directed eastwards and the other is directed northwards.
What is the direction of a single force that will balance these two forces?
A towards the north-east **C** towards the south-east
B towards the north-west **D** towards the south-west
- 91 9702/13/O/N/13/Q4**
The spring constant k of a coiled wire spring is given by the equation $k = \frac{Gr^4}{4nR^3}$
where r is the radius of the wire, n is the number of turns of wire and R is the radius of each of the turns of wire. The quantity G depends on the material from which the wire is made.
What is a suitable unit for G ?
A Nm^{-2} **B** Nm^{-1} **C** Nm **D** Nm^2
- 92 9702/11/M/J/14/Q1**
Which pair of units contains one derived unit and one SI base unit?
A ampere coulomb **C** metre second
B kilogram kelvin **D** newton pascal
- 93 9702/11/M/J/14/Q2**
What is equivalent to 2000 microvolts?
A $2\mu\text{JC}^{-1}$ **B** 2mV **C** 2pV **D** 2000mV
- 94 9702/11/M/J/14/Q3**
The speed v of a liquid leaving a tube depends on the change in pressure ΔP and the density ρ of the liquid. The speed is given by the equation
$$v = k \left(\frac{\Delta P}{\rho} \right)^n$$

where k is a constant that has no units.
What is the value of n ?
A $\frac{1}{2}$ **B** 1 **C** $\frac{3}{2}$ **D** 2
- 95 9702/12/M/J/14/Q1***
The maximum theoretical power P of a wind turbine is given by the equation $P = k\rho A v^n$
where ρ is the density of air, A is the area swept by the turbine blades, v is the speed of the air and k is a constant with no units.
What is the value of n ?
A 1 **B** 2 **C** 3 **D** 4
- 96 9702/12/M/J/14/Q2**
What is the unit of resistance when expressed in SI base units?
A $\text{kg m}^2 \text{s}^{-2} \text{A}^{-1}$ **B** $\text{kg m}^2 \text{s}^{-3} \text{A}^{-2}$ **C** $\text{kg m s}^{-2} \text{A}^{-1}$ **D** $\text{kg m s}^{-3} \text{A}^{-1}$
- 97 9702/12/O/N/11/Q1**
Which quantity can be measured in electronvolts (eV)?
A electric charge **B** electric potential **C** energy **D** power
- 98 9702/13/M/J/14/Q2**
The unit of specific heat capacity is $\text{J kg}^{-1} \text{K}^{-1}$.
What is its equivalent in terms of SI base units?
A $\text{kg}^{-1} \text{m}^2 \text{K}^{-1}$ **B** $\text{m s}^{-1} \text{K}^{-1}$ **C** $\text{m s}^{-2} \text{K}^{-1}$ **D** $\text{m}^2 \text{s}^{-2} \text{K}^{-1}$

99 9702/13/M/J/14/Q3*

What is the vertical component of this displacement vector?

- A 3.0 km B 3.8 km
C 4.0 km D 5.0 km



100 9702/12/O/N/14/Q1

A 0.10 kg mass is taken to Mars and then weighed on a spring balance and on a lever balance. The acceleration due to gravity on Mars is 38% of its value on Earth.

What are the readings on the two balances on Mars? (Assume that on Earth $g = 10 \text{ m s}^{-2}$.)

	spring balance / N	lever balance / kg
A	0.38	0.038
B	0.38	0.10
C	1.0	0.038
D	1.0	0.10

101 9702/11/O/N/14/Q2

What is equivalent to the unit of electric field strength?

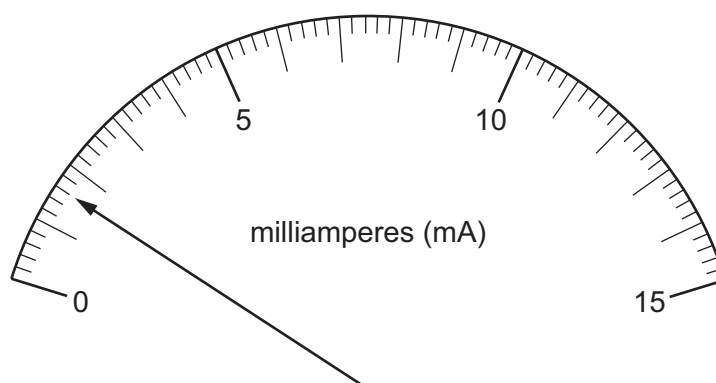
- A J C m^{-1} B N s A^{-1} C $\text{kg m s}^{-3} \text{ A}^{-1}$ D $\text{kg m}^3 \text{ s}^{-3} \text{ A}^{-1}$

102 9702/11/O/N/14/Q3

The diagram shows the reading on an analogue ammeter.

Which digital ammeter reading is the same as the reading on the analogue ammeter?

	display units	display reading
A	μA	1600
B	μA	160
C	mA	16.0
D	A	1.60



103 9702/13/O/N/14/Q1

When the brakes are applied on a vehicle moving at speed v , the distance d moved by the vehicle in coming to rest is given by the expression

$$d = kv^2$$

where k is a constant.

What is the unit of k expressed in SI base units?

- A $\text{m}^{-1} \text{s}^2$ B m s^{-2} C $\text{m}^2 \text{s}^{-2}$ D $\text{m}^{-1} \text{s}$

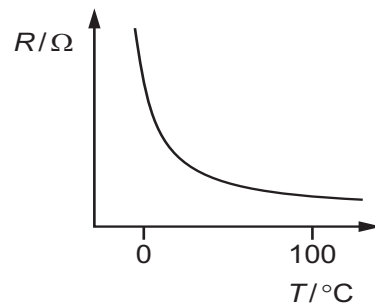
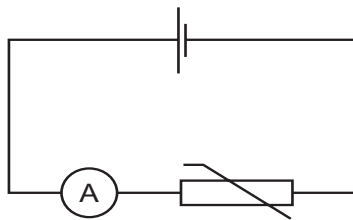
104 9702/13/O/N/14/Q2

Which list contains one vector quantity and two scalar quantities?

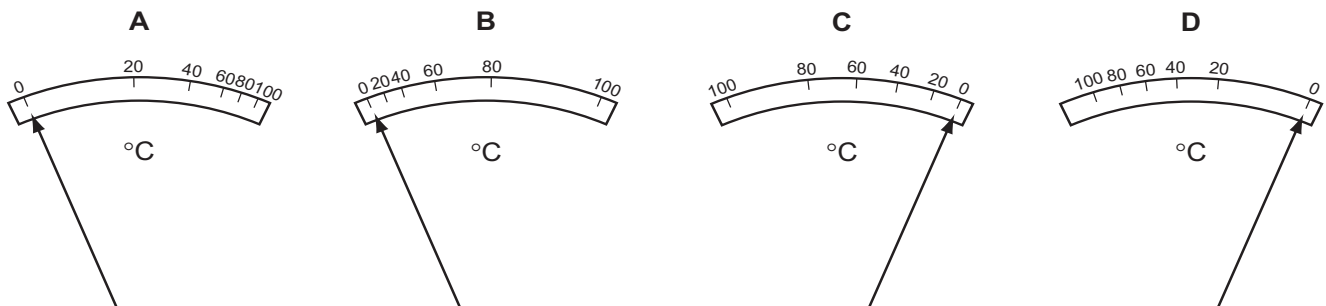
- A** displacement, weight, velocity **C** momentum, mass, speed
B force, acceleration, time **D** work, density, energy

105 9702/13/O/N/14/Q3

In the circuit shown, an analogue ammeter is to be recalibrated as a thermometer. The graph shows how the resistance R of the thermistor changes with temperature T .



Which diagram could represent the temperature scale on the ammeter?



106 9702/11/M/J/15/Q1

Which is an SI base unit?

- A** current **B** gram **C** kelvin **D** volt

107 9702/11/M/J/15/Q2

Which pair contains one vector and one scalar quantity?

- A** displacement acceleration **C** momentum velocity
B force kinetic energy **D** power speed

108 9702/11/M/J/15/Q3

When a constant braking force is applied to a vehicle moving at speed v , the distance d moved by the vehicle in coming to rest is given by the expression

$$d = kv^2$$

where k is a constant.

When d is measured in metres and v is measured in metres per second, the constant has a value of k_1 .

What is the value of the constant when the distance is measured in metres, and the speed is measured in kilometres per hour?

- A** $0.0772k_1$ **B** $0.278k_1$ **C** $3.60k_1$ **D** $13.0k_1$

109 9702/12/M/J/15/Q1

Which definition is correct and uses only quantities rather than units?

- A** Density is mass per cubic metre. **B** Potential difference is energy per unit current.
C Pressure is force per unit area. **D** Speed is distance travelled per second.

110 9702/12/M/J/15/Q2

The average kinetic energy E of a gas molecule is given by the equation $E = \frac{3}{2} kT$ where T is the absolute (kelvin) temperature.

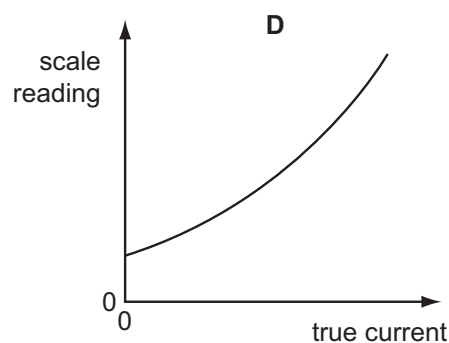
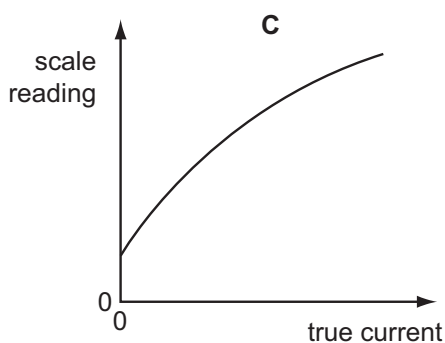
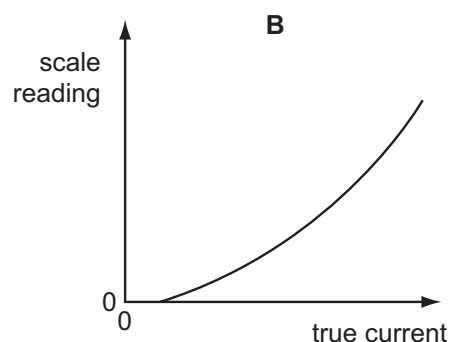
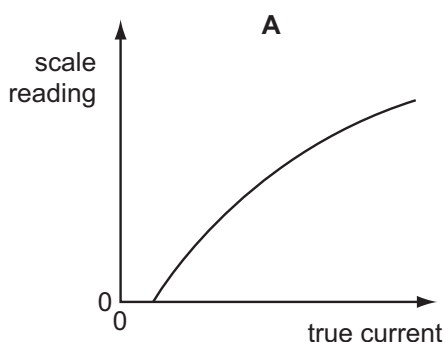
What are the SI base units of k ?

- A** $\text{kg}^{-1} \text{m}^{-1} \text{s}^2 \text{K}$ **B** $\text{kg}^{-1} \text{m}^{-2} \text{s}^2 \text{K}$ **C** $\text{kg m s}^{-2} \text{K}^{-1}$ **D** $\text{kg m}^2 \text{s}^{-2} \text{K}^{-1}$

111 9702/12/M/J/15/Q3

An analogue ammeter has a pointer which moves over a scale. Following prolonged use, the pointer does not return fully to zero when the current is turned off and the meter has become less sensitive at higher currents than it is at lower currents.

Which diagram best represents the calibration graph needed to obtain an accurate current reading?

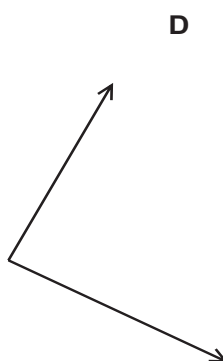
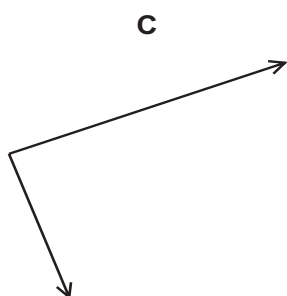
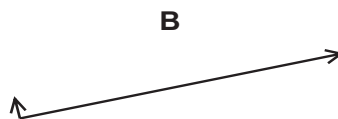
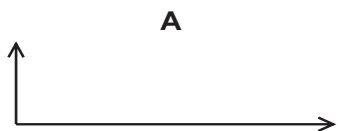


112 9702/12/M/J/15/Q4

The arrow represents the vector R.



Which diagram does **not** represent R as two perpendicular components?



113 9702/13/M/J/15/Q1

Which statement includes a correct unit?

- | | |
|---------------------------|-----------------------------|
| A energy = 7.8 N s | C momentum = 6.2 N s |
| B force = 3.8 N s | D torque = 4.7 N s |

114 9702/13/M/J/15/Q2

What is the joule (J) in SI base units?

- A** kg m s^{-1} **B** $\text{kg m}^2 \text{s}^{-1}$ **C** kg m s^{-2} **D** $\text{kg m}^2 \text{s}^{-2}$

115 9702/13/M/J/15/Q3

The speed of an aeroplane in still air is 200 km h^{-1} . The wind blows from the west at a speed of 85.0 km h^{-1} .

In which direction must the pilot steer the aeroplane in order to fly due north?

- | | |
|-------------------------------------|-------------------------------------|
| A 23.0° east of north | C 25.2° east of north |
| B 23.0° west of north | D 25.2° west of north |

116 9702/11/M/J/17/Q2

A particle travels in a straight line with speed v .

The particle slows down and changes direction. The new speed of the particle is $\frac{v}{2}$.

The new velocity has a component of $\frac{v}{4}$ in the same direction as the initial path of the particle.

Through which angle has the particle turned?

- A** 27° **B** 30° **C** 45° **D** 60°

117 9702/12/M/J/17/Q3

What correctly expresses the volt in terms of SI base units?

- A $A\Omega$ B WA^{-1} C $kg\,m^2\,s^{-1}\,A^{-1}$ D $kg\,m^2\,s^{-3}\,A^{-1}$

118 9702/11/O/N/17/Q1

Which SI unit, expressed in base units, is **not** correct?

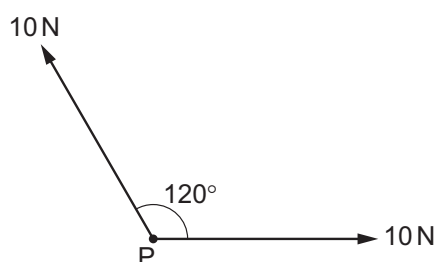
- A unit of force, $kg\,m\,s^{-2}$ C unit of pressure, $kg\,m^{-2}\,s^{-2}$
 B unit of momentum, $kg\,m\,s^{-1}$ D unit of work, $kg\,m^2\,s^{-2}$

119 9702/11/O/N/17/Q2

Two forces, each of 10 N, act at a point P as shown. The angle between the directions of the forces is 120° .

What is the magnitude of the resultant force?

- A 5 N C 17 N
 B 10 N D 20 N



120 9702/11/O/N/17/Q6

In still air, a bird can fly at a speed of $10\,m\,s^{-1}$. The wind is blowing from the east at $8.0\,m\,s^{-1}$.

In which direction must the bird fly in order to travel to a destination that is due north of the bird's current location?

- A 37° east of north C 53° east of north
 B 37° west of north D 53° west of north

121 9702/12/O/N/17/Q1

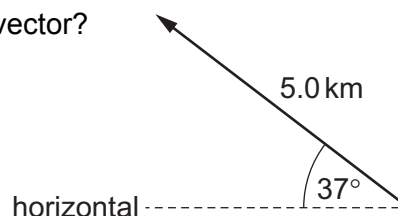
Which pair of units are **not** the same when expressed in SI base units?

- A $m\,s^{-2}$ and $N\,kg^{-1}$ C Pa and $N\,m^{-2}$
 B Ns and $kg\,m\,s^{-1}$ D $V\,m^{-2}$ and $N\,C^{-1}$

122 9702/12/O/N/17/Q2

What is the vertical component of this displacement vector?

- A 3.0 km B 3.8 km
 C 4.0 km D 5.0 km



123 9702/12/O/N/17/Q3

The units of specific heat capacity are $J\,kg^{-1}\,K^{-1}$

What are the SI base units of specific heat capacity?

- A $m\,s^{-2}\,K^{-1}$ B $m\,s^{-1}\,K^{-1}$ C $m^2\,s^{-2}\,K^{-1}$ D $m^2\,s^{-1}\,K^{-1}$

124 9702/12/O/N/17/Q4

A quantity y is to be determined from the equation shown. $y = \frac{px}{q^2}$

The percentage uncertainties in p , x and q are shown.

What is the percentage uncertainty in y ?

- A** 0.5% **B** 0.75%
C 12% **D** 16%

	percentage uncertainty
p	6%
x	2%
q	4%

125 9702/13/O/N/17/Q1

How many cubic nanometres, nm^3 , are in a cubic micrometre, μm^3 ?

- A** 10^3 **B** 10^6 **C** 10^9 **D** 10^{12}

126 9702/13/O/N/17/Q2

The maximum theoretical power P of a wind turbine is given by the equation $P = k\rho Av^n$

where ρ is the density of air, A is the area swept by the turbine blades, v is the speed of the air and k is a constant with no units.

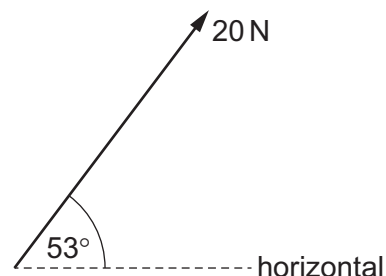
What is the value of n ?

- A** 1 **B** 2 **C** 3 **D** 4

127 9702/13/O/N/17/Q3

What is the horizontal component of the force shown?

- A** 12 N
B 16 N
C 20 N
D 27 N



128 9702/11/M/J/18/Q1

What is a unit for stress?

- A** $\text{kg m}^{-1} \text{s}^{-2}$ **B** $\text{kg m}^{-2} \text{s}^{-2}$ **C** Nm^{-1} **D** Nm

129 9702/11/M/J/18/Q2

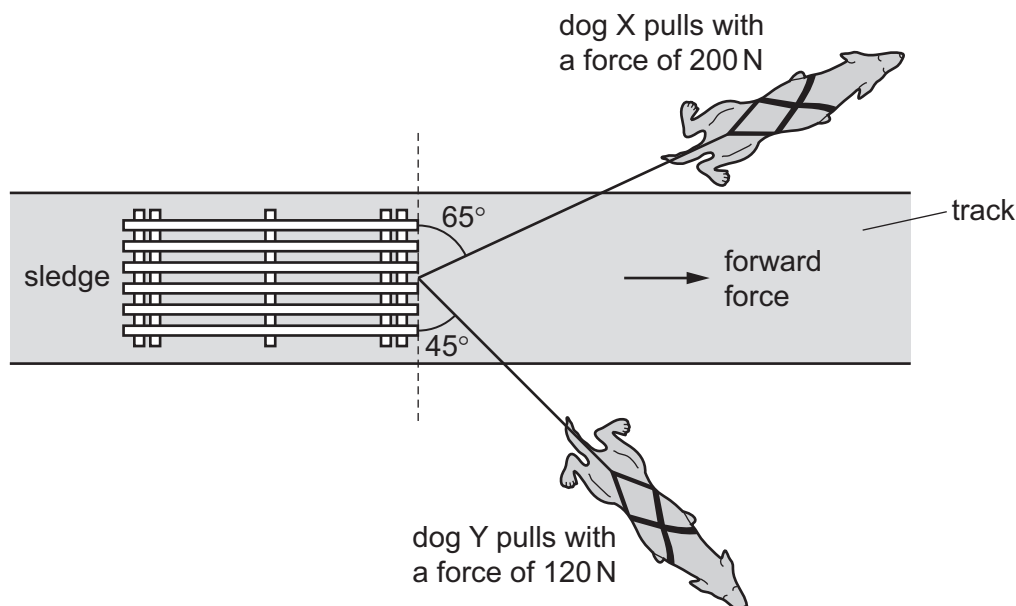
Physical quantities can be classed as vectors or as scalars.

Which pair of quantities consists of two vectors?

- A** kinetic energy and force **C** velocity and electric field strength
B momentum and time **D** weight and temperature

130 9702/11/M/J/18/Q3

Two dogs pull a sledge along an icy track, as shown.



Dog X pulls with a force of 200 N at an angle of 65° to the front edge of the sledge. Dog Y pulls with a force of 120 N at an angle of 45° to the front edge of the sledge.

What is the resultant forward force on the sledge exerted by the two dogs?

- A** 80 N **B** 170 N **C** 270 N **D** 320 N

131 9702/12/M/J/18/Q1

A sheet of gold leaf has a thickness of 0.125 μm. A gold atom has a radius of 174 pm. Approximately how many layers of atoms are there in the sheet?

- A** 4 **B** 7 **C** 400 **D** 700

132 9702/12/M/J/18/Q2

The drag coefficient C_d is a number with no units. It is used to compare the drag on different cars at different speeds. C_d is given by the equation

$$C_d = \frac{2F}{v^n \rho A}$$

where F is the drag force on the car, ρ is the density of the air, A is the cross-sectional area of the car and v is the speed of the car.

What is the value of n ?

- A** 1 **B** 2 **C** 3 **D** 4

133 9702/13/M/J/18/Q1

What is the best way of describing a physical quantity?

- A a quantity with a magnitude and a direction but no unit
- B a quantity with a magnitude and a unit
- C a quantity with a magnitude but no direction
- D a quantity with a unit but no magnitude

134 9702/13/M/J/18/Q2

Which pair includes a vector quantity and a scalar quantity?

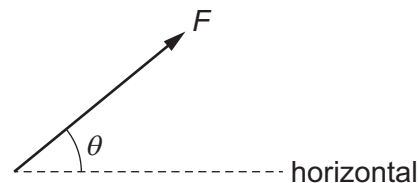
- A displacement and acceleration
- B force and kinetic energy
- C power and speed
- D work and potential energy

135 9702/13/M/J/18/Q3

A force F acts at an angle θ to the horizontal.

What are the horizontal and the vertical components of the force?

	horizontal component	vertical component
A	$F \cos \theta$	$F \cos (90^\circ - \theta)$
B	$F \cos \theta$	$F \sin (90^\circ - \theta)$
C	$F \sin \theta$	$F \cos \theta$
D	$F \sin \theta$	$F \cos (90^\circ - \theta)$



136 9702/12/F/M/18/Q1

Which unit is equivalent to the coulomb?

- A ampere per second
- B joule per volt
- C watt per ampere
- D watt per volt

137 9702/12/F/M/18/Q2

Which row shows a quantity and an **incorrect** unit?

	quantity	unit
A	efficiency	no unit
B	moment of force	Nm^{-1}
C	momentum	Ns
D	work done	J

138 9702/12/F/M/18/Q3

Two forces of equal magnitude are represented by two coplanar vectors. One is directed towards the east and the other is directed towards the north.

What is the direction of a single force that will balance these two forces?

- A towards the north-east
- B towards the north-west
- C towards the south-east
- D towards the south-west

139 9702/11/O/N/18/Q1

The radius of the Earth is approximately $6.4 \times 10^6 \text{ m}$, and the radius of the Moon is approximately $1.7 \times 10^6 \text{ m}$. A student wishes to build a scale model of the Solar System in the classroom, using a football of radius 0.12 m to represent the Earth.

Which object would best represent the Moon?

- A** basketball **C** golf ball
B cherry **D** tennis ball

140 9702/11/O/N/18/Q2

When a beam of light is incident on a surface, it delivers energy to the surface. The intensity of the beam is defined as the energy delivered per unit area per unit time.

What is the unit of intensity, expressed in SI base units?

- A** $\text{kg m}^{-2} \text{ s}^{-1}$ **B** $\text{kg m}^2 \text{ s}^{-3}$ **C** kg s^{-2} **D** kg s^{-3}

141 9702/11/O/N/18/Q3

A ship is travelling with a velocity of 8.0 km h^{-1} in a direction 30° east of north.

What are the components of the ship's velocity in the east and north directions?

	component of velocity in east direction $/\text{km h}^{-1}$	component of velocity in north direction $/\text{km h}^{-1}$
A	4.0	4.0
B	4.0	6.9
C	4.6	6.9
D	6.9	4.0

142 9702/12/O/N/18/Q2

What is the unit of resistance when expressed in SI base units?

- A** $\text{kg m}^2 \text{ s}^{-2} \text{ A}^{-1}$ **C** $\text{kg m s}^{-2} \text{ A}^{-1}$
B $\text{kg m}^2 \text{ s}^{-3} \text{ A}^{-2}$ **D** $\text{kg m s}^{-3} \text{ A}^{-1}$

143 9702/12/O/N/18/Q3

Which list contains both scalar and vector quantities?

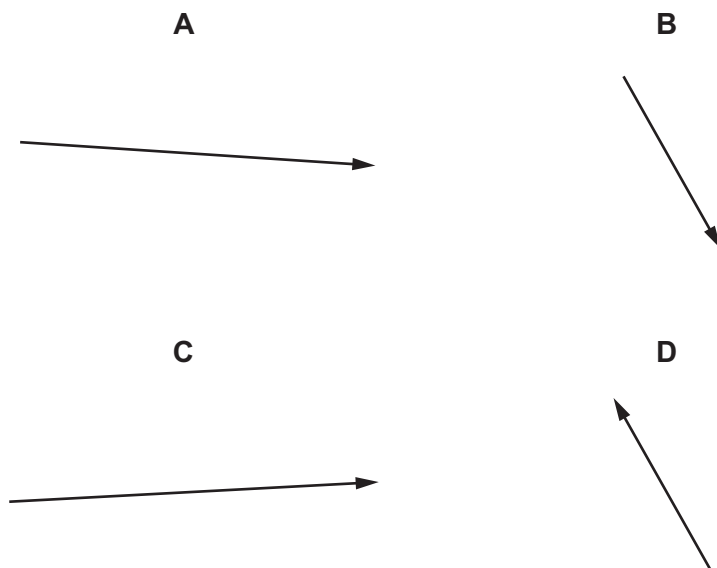
- A** acceleration, momentum, velocity, weight
B area, current, force, work
C distance, kinetic energy, power, pressure
D mass, temperature, time, speed

144 9702/12/O/N/18/Q4

Vectors P and Q are drawn to scale.



Which diagram represents the vector $(P + Q)$?



145 9702/13/O/N/18/Q2

Three of these quantities have the same unit.

Which quantity has a different unit?

- | | |
|--|-------------------------------------|
| A $\frac{\text{energy}}{\text{distance}}$ | C power $\times$ time |
| B force | D rate of change of momentum |

146 9702/13/O/N/18/Q3

Which group of quantities contains only vectors?

- A** acceleration, displacement, speed
- B** acceleration, work, electric field strength
- C** displacement, force, velocity
- D** power, electric field strength, force

147 9702/11/M/J/19/Q1

Which unit can be expressed in base units as $\text{kg m}^2 \text{s}^{-2}$?

- A** joule **C** pascal
B newton **D** watt

148 9702/11/M/J/19/Q2

The luminosity L of a star is given by

$$L = 4\pi r^2 \sigma T^4$$

where

r is the radius of the star,

T is the temperature of the star and

σ is a constant with units $\text{W m}^{-2} \text{K}^{-4}$.

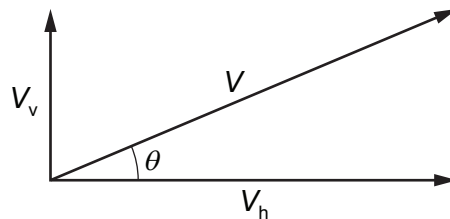
What are the SI base units of L ?

- A** $\text{kg m}^2 \text{s}^{-1}$ **B** $\text{kg m}^2 \text{s}^{-2}$ **C** $\text{kg m}^2 \text{s}^{-3}$ **D** $\text{kg m}^2 \text{s}^{-4}$

149 9702/11/M/J/19/Q3

A particle has velocity V at an angle θ to the horizontal.

The components of the particle's velocity are V_v upwards in the vertical direction and V_h to the right in the horizontal direction, as shown.



What are expressions for the magnitude of V and for the angle θ ?

	magnitude of V	θ
A	$\sqrt{(V_v^2 + V_h^2)}$	$\tan^{-1} \left(\frac{V_h}{V_v} \right)$
B	$\sqrt{(V_v^2 + V_h^2)}$	$\tan^{-1} \left(\frac{V_v}{V_h} \right)$
C	$\sqrt{(V_v^2 - V_h^2)}$	$\tan^{-1} \left(\frac{V_h}{V_v} \right)$
D	$\sqrt{(V_v^2 - V_h^2)}$	$\tan^{-1} \left(\frac{V_v}{V_h} \right)$

150 9702/12/M/J/19/Q1

What is equivalent to 2000 microvolts?

- A** $2\mu\text{J C}^{-1}$ **B** 2 mV **C** 2 pV **D** 2000 mV

151 9702/12/M/J/19/Q2

What is the number of SI base units required to express electric field strength and power?

	electric field strength	power
A	3	3
B	3	2
C	4	2
D	4	3

152 9702/12/M/J/19/Q3

The Planck constant h has SI units J s.

Which equation could be used to calculate the Planck constant?

A $h = \frac{DE}{v}$ where D is distance, E is energy and v is velocity

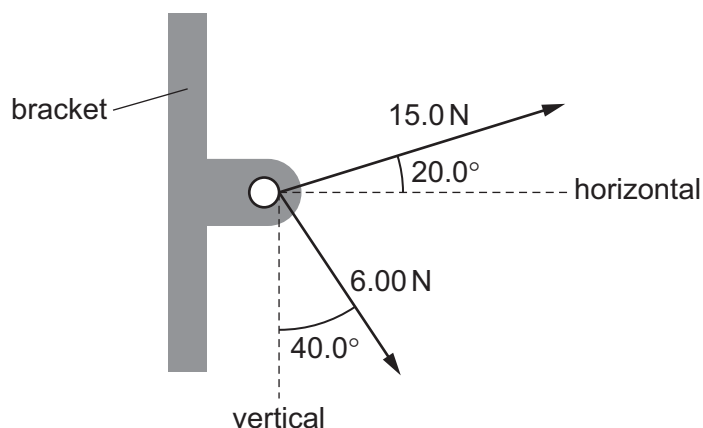
B $h = \frac{v}{D}$ where v is velocity and D is distance

C $h = \frac{1}{4\pi E}$ where E is electric field strength

D $h = \frac{Fr^2}{m}$ where F is force, r is radius and m is mass

153 9702/12/M/J/19/Q4

Two cables are attached to a bracket and exert forces as shown.



What are the magnitudes of the horizontal and vertical components of the resultant of the two forces?

	horizontal component / N	vertical component / N
A	9.73	0.534
B	9.73	10.2
C	18.0	0.534
D	18.0	10.2

154 9702/13/M/J/19/Q1

Which is an SI base unit?

- | | |
|------------------|-----------------|
| A current | C kelvin |
| B gram | D volt |

155 9702/13/M/J/19/Q2

-3

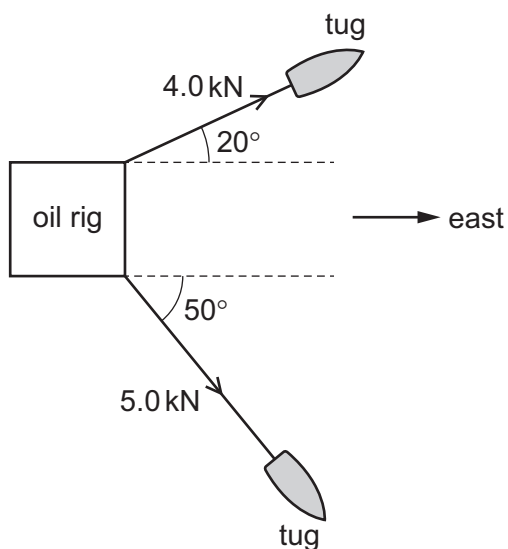
Osmium, a naturally occurring element, has a density of 23 000 kg m⁻³.

What is also a value of the density of osmium?

- | | |
|--|--|
| A $2.3 \times 10^4 \mu\text{g cm}^{-3}$ | C 2.3 kg cm^{-3} |
| B $2.3 \times 10^4 \text{ g cm}^{-3}$ | D $2.3 \times 10^{-2} \text{ kg cm}^{-3}$ |

156 9702/13/M/J/19/Q3

Two tugs are towing an oil rig as shown.



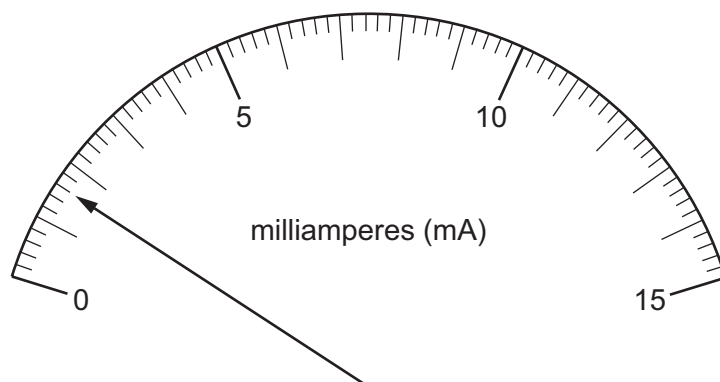
The tensions in the towing cables are 4.0 kN and 5.0 kN.

What is the total force acting on the rig due to the cables, in the direction to the east?

- A** 3.1 kN **B** 5.2 kN **C** 7.0 kN **D** 7.3 kN

157 9702/13/M/J/19/Q5

The diagram shows the reading on an analogue ammeter.

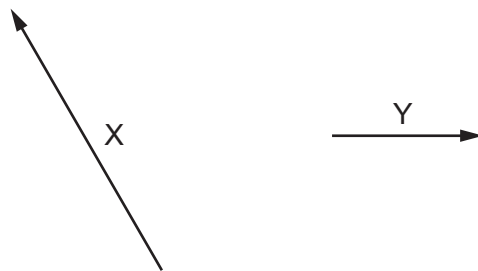


Which digital ammeter reading is the same as the reading on the analogue ammeter?

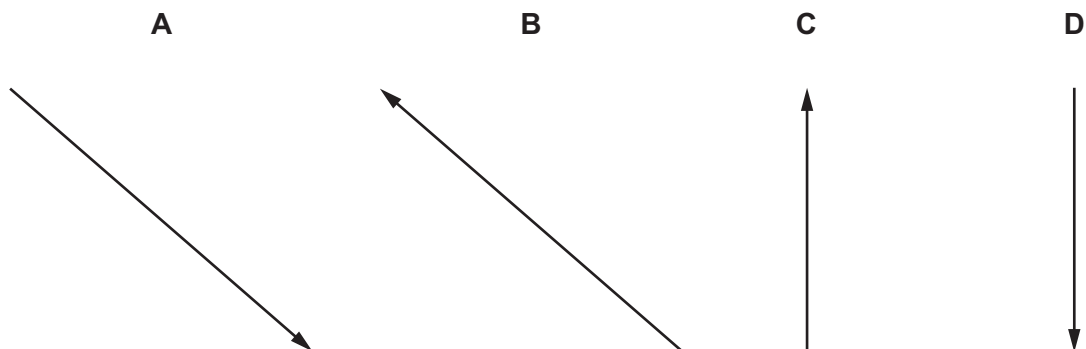
	display units	display reading
A	μA	1600
B	μA	160
C	mA	16.0
D	A	1.60

158 9702/11/O/N/19/Q3

The diagram shows two vectors X and Y , drawn to scale.



If $X = Y - Z$, which diagram best represents the vector Z ?



159 9702/12/O/N/19/Q2

Which expression gives an SI base quantity?

- A** charge per unit time
- B** force per unit area
- C** mass per unit volume
- D** work done per unit distance

160 9702/12/O/N/19/Q3

Which list contains only scalar quantities?

- A** area, length, displacement
- B** kinetic energy, speed, power
- C** potential energy, momentum, time
- D** velocity, distance, temperature

161 9702/13/O/N/19/Q1

Which quantity with its unit is correct?

- A acceleration of a bicycle = 1.4 m s^{-1}
- B electric current in a lamp = 0.25 A s^{-1}
- C electric potential difference across a battery = 8.0 J C^{-1}
- D kinetic energy of a car = 4500 N m^{-1}

162 9702/13/O/N/19/Q2

Which two units are **not** equivalent to each other?

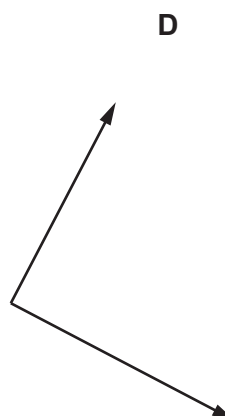
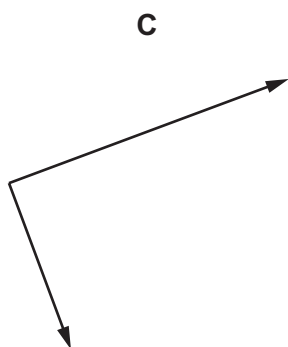
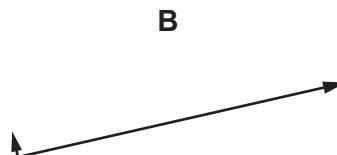
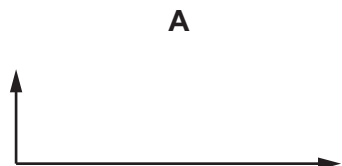
- A N m and $\text{kg m}^2 \text{ s}^{-2}$
- B N s and kg m s^{-1}
- C J s^{-1} and $\text{kg m}^2 \text{ s}^{-3}$
- D Pa and kg m s^{-2}

163 9702/13/O/N/19/Q3

The arrow represents a vector R.



Which diagram does **not** represent R as two perpendicular components?



164. 9702/12/F/M/20/Q1

The table shows some measurable quantities.

Which row gives the correct order of magnitude of the measurable quantity in the stated unit?

	measurable quantity	order of magnitude	unit
A	mass of a coin	10^{-4}	kg
B	thickness of a sheet of paper	10^{-2}	m
C	weight of an apple	10^0	N
D	temperature of a person's body	10^1	K

165. 9702/12/F/M/20/Q2

A byte (b) comprises 8 bits.

How many bits are there in 1 terabyte (1Tb)?

- A** 1×10^9 **B** 8×10^9 **C** 1×10^{12} **D** 8×10^{12}

166. 9702/12/F/M/20/Q3

Which pair of quantities contains both a scalar **and** a vector?

- A** acceleration and momentum
B charge and resistance
C kinetic energy and mass
D temperature and velocity

167. 9702/11/M/J/20/Q1

What is a reasonable estimate of the kinetic energy of a car travelling at a speed of 30 m s^{-1} ?

- A** 10^2 J **B** 10^4 J **C** 10^6 J **D** 10^8 J

168. 9702/11/M/J/20/Q2

The frequency f of vibration of a mass m supported by a spring with spring constant k is given by the equation

$$f = Cm^p k^q$$

where C is a constant with no units.

What are the values of p and q ?

	p	q
A	$-\frac{1}{2}$	$-\frac{1}{2}$
B	$-\frac{1}{2}$	$\frac{1}{2}$
C	$\frac{1}{2}$	$-\frac{1}{2}$
D	$\frac{1}{2}$	$\frac{1}{2}$

169. 9702/11/M/J/20/Q3

The power produced by a force moving an object is given by the equation shown.

$$\text{power} = \frac{\text{work}}{\text{time}} = \frac{\text{force} \times \text{displacement}}{\text{time}}$$

Which quantities are scalars and which are vectors?

	scalars	vectors
A	displacement, time	force, power
B	power, work	displacement, force
C	power, force	displacement, work
D	work, time	power, displacement

170. 9702/12/M/J/20/Q1

What is a reasonable estimate of the mass of a raindrop?

- A** 10^1 kg **B** 10^{-1} kg **C** 10^{-3} kg **D** 10^{-5} kg

171. 9702/12/M/J/20/Q2

Which quantity is a scalar?

- A** acceleration
B force
C kinetic energy
D momentum

172. 9702/13/M/J/20/Q1

A man is running a race in a straight line.

What is an approximate value of his kinetic energy?

- A** 10 J **B** 100 J **C** 1000 J **D** 10 000 J

173. 9702/13/M/J/20/Q2

A sample of gas has a mass of $4.8 \mu\text{g}$ and occupies a volume of 1.2 dm^3 .

What is the density of the sample of gas?

- A** $4.0 \times 10^{-3} \text{ kg m}^{-3}$
B $4.0 \times 10^{-5} \text{ kg m}^{-3}$
C $4.0 \times 10^{-6} \text{ kg m}^{-3}$
D $4.0 \times 10^{-8} \text{ kg m}^{-3}$

174. 9702/13/M/J/20/Q3

Which characteristics are possessed by a vector quantity but **not** by a scalar quantity?

- A** direction only
- B** magnitude and direction
- C** magnitude and unit
- D** unit only

175. 9702/13/M/J/20/Q4

A circuit is set up in order to determine the resistance of a 12V, 1.2W lamp when operating normally. An analogue ammeter and an analogue voltmeter are used.

Which ranges for the meters would be most suitable?

	ammeter range /A	voltmeter range /V
A	0–0.5	0–20
B	0–0.5	0–100
C	0–10	0–20
D	0–10	0–100

176. 9702/11/O/N/20/Q1

Which quantity is a physical quantity?

- A** atomic number
- B** efficiency
- C** number density of charge carriers
- D** strain

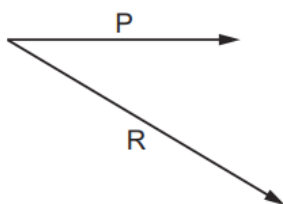
177. 9702/11/O/N/20/Q2

Which time interval is the shortest?

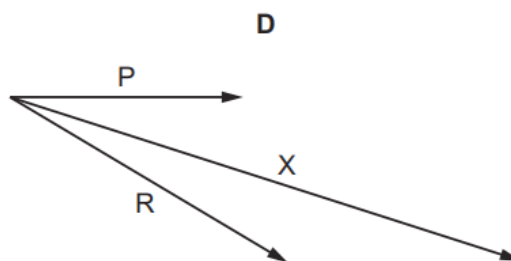
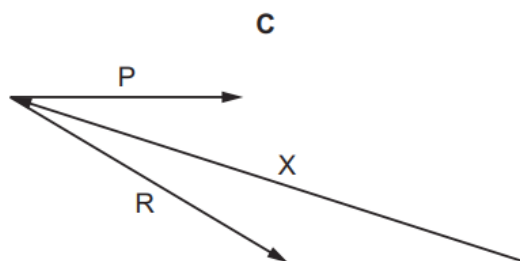
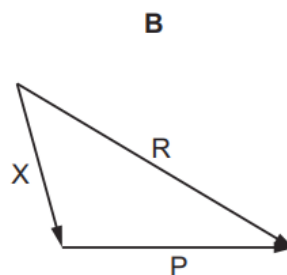
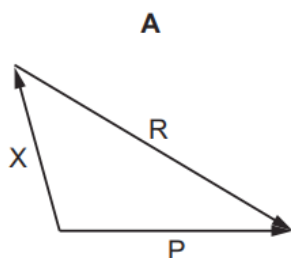
- A** 0.05 ms **B** 50 ns **C** 500 000 ps **D** 0.5 μ s

178. 9702/11/O/N/20/Q3

P and R are coplanar vectors.



If $X = P - R$, which diagram best represents vector X?



179. 9702/12/O/N/20/Q2

The speed v of waves on a stretched wire is given by the equation

$$v = T^p \mu^q$$

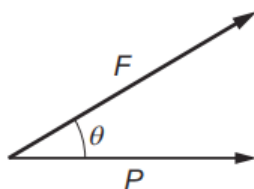
where T is the tension in the wire and μ is the mass per unit length of the wire.

What are the values of p and q ?

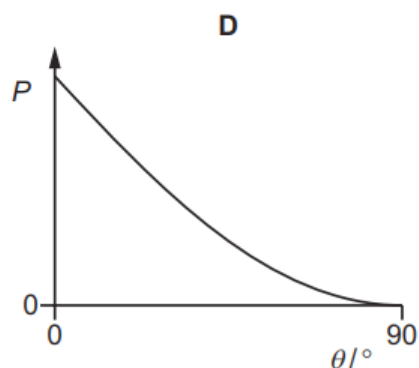
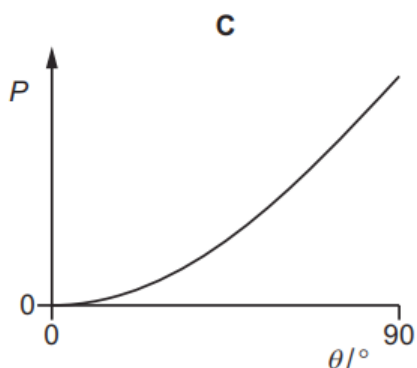
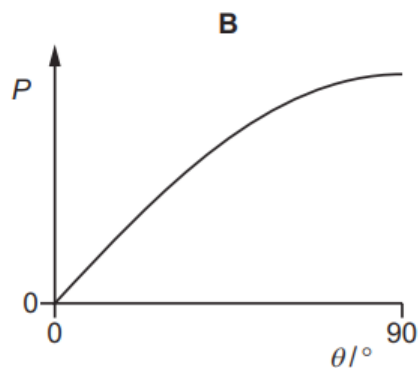
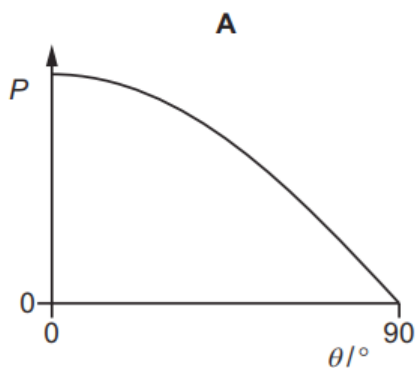
	p	q
A	$-\frac{1}{2}$	$-\frac{1}{2}$
B	$-\frac{1}{2}$	$\frac{1}{2}$
C	$\frac{1}{2}$	$-\frac{1}{2}$
D	$\frac{1}{2}$	$\frac{1}{2}$

180. 9702/12/O/N/20/Q3

The diagram shows a force F . P is the horizontal component of F , at an angle θ to F .



Which graph best shows the variation with θ of the magnitude of P ?



181. 9702/13/O/N/20/Q1

What is a reasonable estimate of the volume of a fully inflated standard football?

- A** 600 cm^3 **B** 6000 cm^3 **C** $60\,000 \text{ cm}^3$ **D** $600\,000 \text{ cm}^3$

182. 9702/13/O/N/20/Q2

What is **not** an SI base unit?

- A** coulomb
B kelvin
C kilogram
D second

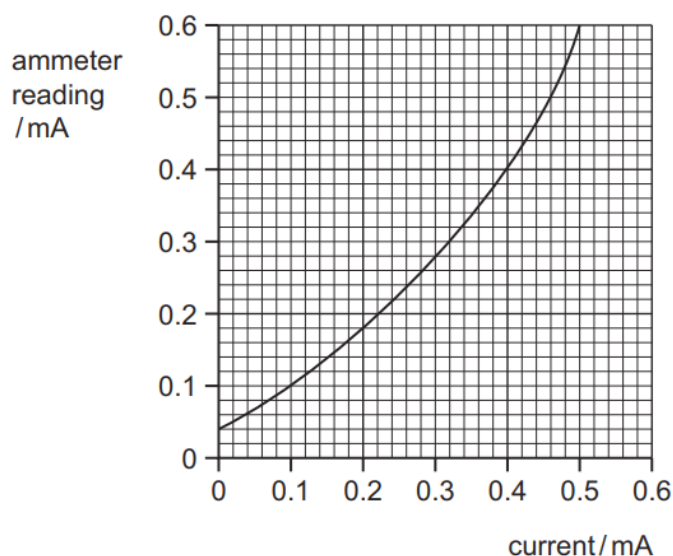
183. 9702/13/O/N/20/Q3

Which two quantities are both vector quantities?

- A** displacement and distance
- B** force and momentum
- C** torque and time
- D** weight and pressure

184. 9702/13/O/N/20/Q4

A calibration curve is shown for an ammeter whose scale is inaccurate.



Two readings taken on the meter at different times during an experiment are 0.13 mA and 0.47 mA.

By how much did the current really increase between taking the two readings?

- A** 0.30 mA **B** 0.34 mA **C** 0.40 mA **D** 0.44 mA

185. 9702/12/F/M/21/Q1

What is a reasonable estimate for the density of sand?

- A** $2 \times 10^2 \text{ g cm}^{-3}$
- B** $2 \times 10^3 \text{ g cm}^{-3}$
- C** $2 \times 10^1 \text{ kg m}^{-3}$
- D** $2 \times 10^3 \text{ kg m}^{-3}$

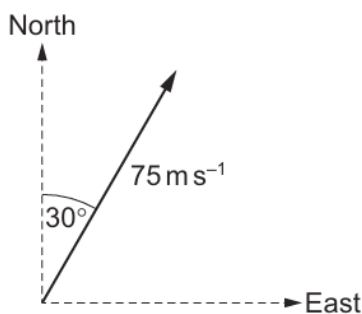
186. 9702/12/F/M/21/Q2

Which physical quantity could have units of $\text{Ns}^2 \text{ m}^{-1}$?

- A** acceleration
- B** force
- C** mass
- D** momentum

187. 9702/12/F/M/21/Q3

A velocity vector is shown.



What are the components of the velocity vector in the northerly and in the easterly directions?

	component of vector in northerly direction $/\text{m s}^{-1}$	component of vector in easterly direction $/\text{m s}^{-1}$
A	38	38
B	38	65
C	65	38
D	65	65

188. 9702/11/M/J/21/Q1

What is a reasonable estimate of the volume of an adult person?

- A** 0.10 m^3 **B** 0.50 m^3 **C** 1.0 m^3 **D** 2.0 m^3

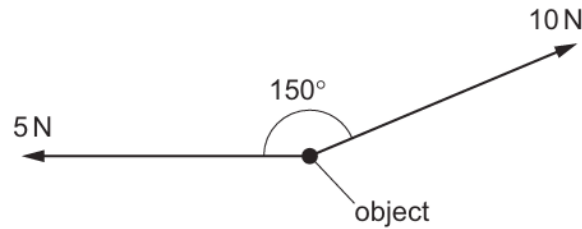
189. 9702/11/M/J/21/Q2

Which combination of units could be used for expressing the power dissipated in a resistor?

- A** newton per second (N s^{-1})
B newton second (Ns)
C newton metre (Nm)
D newton metre per second (Nm s^{-1})

190. 9702/11/M/J/21/Q3

A force of 10 N and a force of 5 N act on an object.



The angle between the forces is 150° .

The resultant force on the object can be resolved into a pair of perpendicular components.

Which row shows numerical expressions for a possible pair of perpendicular components?

	force component / N	force component / N
A	$10 \cos 30^\circ - 5$	$10 \cos 30^\circ$
B	$10 \sin 30^\circ - 5$	$10 \cos 30^\circ$
C	$10 - 5 \cos 30^\circ$	$5 \sin 30^\circ$
D	$10 - 5 \sin 30^\circ$	$5 \cos 30^\circ$

191. 9702/11/M/J/21/Q5

A micrometer screw gauge is used to measure the diameter of a wire.

The reading on the micrometer with the jaws closed is (-0.05 ± 0.02) mm.

The reading with the wire in position between the two jaws is $(+1.03 \pm 0.02)$ mm.

What is the diameter of the wire?

- A** (0.98 ± 0.02) mm
- B** (1.08 ± 0.02) mm
- C** (0.98 ± 0.04) mm
- D** (1.08 ± 0.04) mm

192. 9702/12/M/J/21/Q1

What is **not** a reasonable estimate of the physical property indicated?

- A** 2×10^3 W for the power dissipated by the heating element of an electric kettle
- B** 4×10^2 m³ for the volume of water in a swimming pool
- C** 5×10^5 N s for the momentum of a lorry moving along a road
- D** 6×10^2 N for the weight of a fully grown racehorse

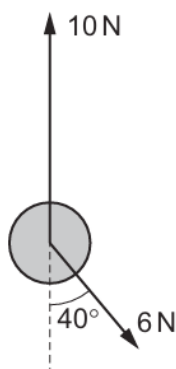
193. 9702/12/M/J/21/Q2

Which quantity could have units of N m V^{-1} ?

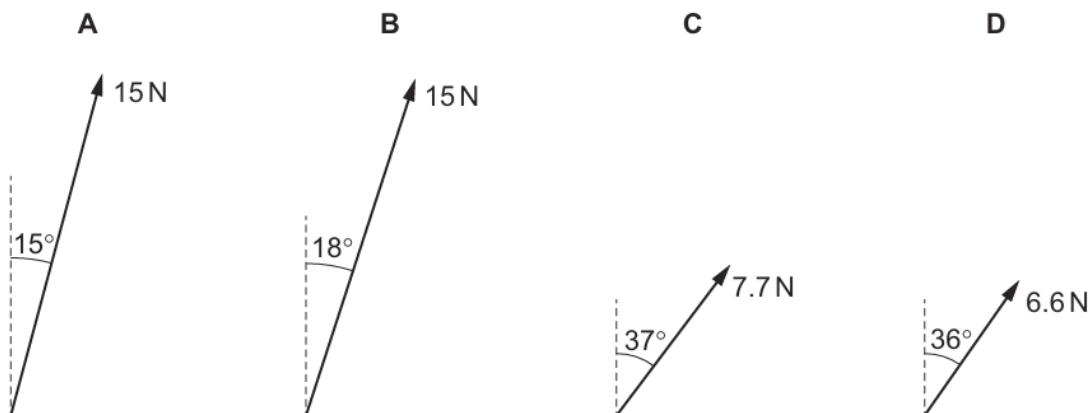
- A acceleration
- B charge
- C current
- D resistance

194. 9702/12/M/J/21/Q3

An object is acted upon by two forces, 10 N in the vertical direction and 6 N at 40° to the vertical, as shown.



What is the resultant force acting on the object?



195. 9702/13/M/J/21/Q1

What is a reasonable estimate of the kinetic energy of an Olympic athlete sprinting in a 100 m race?

- A 40 J
- B 400 J
- C 4000 J
- D 40 000 J

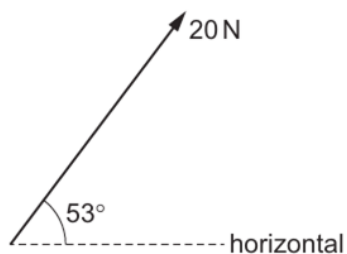
196. 9702/13/M/J/21/Q2

What is a unit of momentum?

- A kg m s^{-2}
- B N s^{-1}
- C N s
- D kg s m^{-1}

197. 9702/13/M/J/21/Q3

What is the horizontal component of the force shown?



- A** 12 N **B** 16 N **C** 25 N **D** 27 N

198. 9702/12/O/N/21/Q1

Which row shows what all physical quantities must have?

	magnitude	direction	unit
A	✓	✓	✓
B	✓	✓	✗
C	✓	✗	✓
D	✗	✗	✓

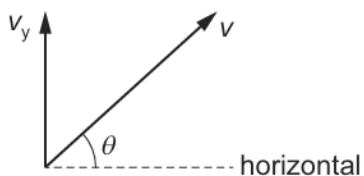
199. 9702/12/O/N/21/Q2

What is an alternative way of expressing an energy of 43 dJ?

- A** $4.3 \times 10^3 \text{ mJ}$
B $4.3 \times 10^3 \text{ MJ}$
C $4.3 \times 10^{-3} \text{ mJ}$
D $4.3 \times 10^{-3} \text{ MJ}$

200. 9702/12/O/N/21/Q3

A tennis ball is hit so that it leaves the racket with velocity v at an angle θ to the horizontal.



The vertical component of the velocity is v_y .

What is the magnitude of the horizontal component of v ?

- A** $v \sin \theta$ **B** $v_y \cos \theta$ **C** $v_y \sin \theta$ **D** $(v^2 - v_y^2)^{\frac{1}{2}}$

201. 9702/11/O/N/21/Q1

What is essential when recording a measurement of a physical quantity?

- A** the measurement has an SI unit
- B** the measurement has a unit and a number
- C** the measurement has a unit given as a base unit
- D** the measurement is from an analogue scale

202. 9702/11/O/N/21/Q2

The mobility μ of electrons travelling through a metal conductor can be calculated using the equation

$$\mu = \left(\frac{e}{m} \right) \tau$$

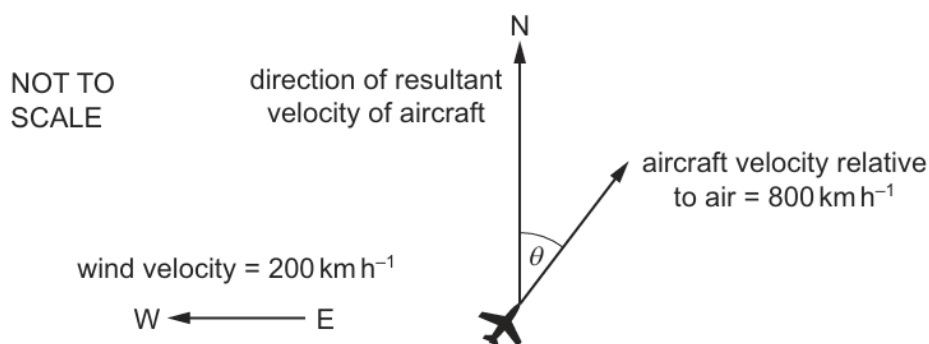
where e is the charge on an electron and m is its mass. The average time between the collisions of an electron with the atoms in the metal is τ .

What are the SI base units of μ ?

- A** A kg^{-1}
- B** $\text{A s}^2 \text{kg}^{-1}$
- C** A s kg^{-1}
- D** $\text{A s}^{-2} \text{kg}^{-1}$

203. 9702/11/O/N/21/Q3

An aircraft heads in a direction at an angle θ east of north with a horizontal velocity relative to the air of 800 km h^{-1} . The wind blows with a horizontal velocity of 200 km h^{-1} from east to west, as shown.



The resultant velocity of the aircraft is in a direction due north.

What is angle θ and what is the magnitude of the resultant velocity?

	$\theta / ^\circ$	resultant velocity / km h^{-1}
A	14	770
B	14	820
C	76	770
D	76	820

204. 9702/13/O/N/21/Q2

What is the unit of resistance when expressed in SI base units?

- A $\text{kg}^{-1} \text{m}^{-2} \text{sA}^2$
- B $\text{kg}^{-1} \text{m}^{-2} \text{s}^3 \text{A}^2$
- C $\text{kg m}^2 \text{s}^{-1} \text{A}^{-2}$
- D $\text{kg m}^2 \text{s}^{-3} \text{A}^{-2}$

205. 9702/13/O/N/21/Q3

Which list consists only of scalar quantities?

- A acceleration, displacement, force, weight
- B density, energy, frequency, velocity
- C distance, pressure, temperature, time
- D momentum, power, volume, wavelength

206. 9702/12/F/M/22/Q1

What could **not** be a measurement of a physical quantity?

- A 10 K
- B $11 \text{ J N}^{-1} \text{ m}^{-1}$
- C $17 \text{ Pa m}^3 \text{ N}^{-1}$
- D 25 T m

207. 9702/12/F/M/22/Q2

A computer memory stick is labelled as having a storage capacity of 128 GB.

The letter B stands for byte, which is a unit.

What is the equivalent storage capacity?

- A $1.28 \times 10^8 \text{ B}$
- B $1.28 \times 10^{11} \text{ B}$
- C $1.28 \times 10^{14} \text{ B}$
- D $1.28 \times 10^{17} \text{ B}$

208. 9702/12/F/M/22/Q4

Which statement about scalar and vector quantities is correct?

- A A scalar quantity has direction but not magnitude.
- B A scalar quantity has magnitude but not direction.
- C A vector quantity has direction but not magnitude.
- D A vector quantity has magnitude but not direction.

209. 9702/11/M/J/22/Q1

Which term represents a physical quantity?

- A metre
- B percentage uncertainty
- C quark flavour
- D spring constant

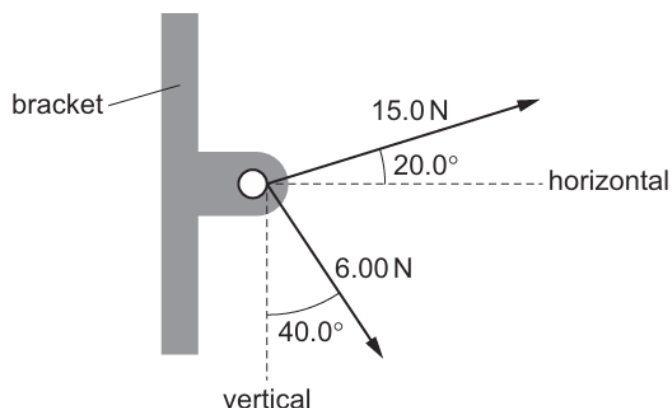
210. 9702/11/M/J/22/Q2

Which two units are identical when expressed in terms of SI base units?

- A JC^{-1} and $\text{kg m}^2 \text{A}^{-1} \text{s}^{-2}$
- B Js and $\text{kg m}^2 \text{s}^{-1}$
- C Nm and $\text{kg m}^3 \text{s}^{-2}$
- D Ns and kg m s^{-3}

211. 9702/11/M/J/22/Q4

Two cables are attached to a bracket and exert forces as shown.



What are the magnitudes of the horizontal and vertical components of the resultant of the two forces?

	horizontal component / N	vertical component / N
A	9.73	0.534
B	9.73	10.2
C	18.0	0.534
D	18.0	10.2

212. 9702/12/M/J/22/Q1

Which estimate is reasonable?

- A $1 \times 10^{-3} \text{ kg}$ for the mass of a grain of sand
- B $1 \times 10^{-2} \text{ m}^3$ for the volume of a tennis ball
- C $1 \times 10^0 \text{ J}$ for the work done lifting an apple from waist height to head height
- D $1 \times 10^4 \text{ W}$ for the power of a light bulb in a house

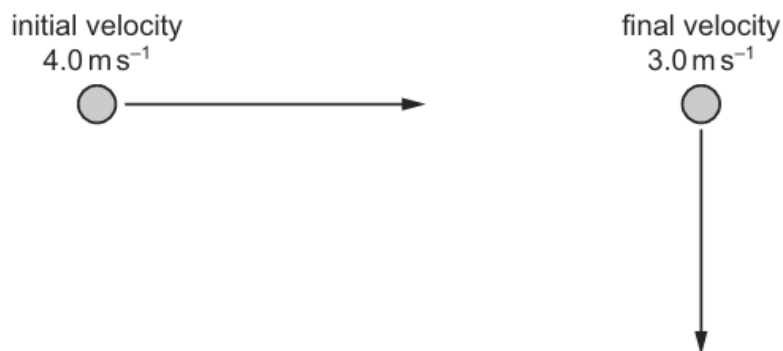
213. 9702/12/M/J/22/Q2

What is the symbol for the SI base unit of temperature?

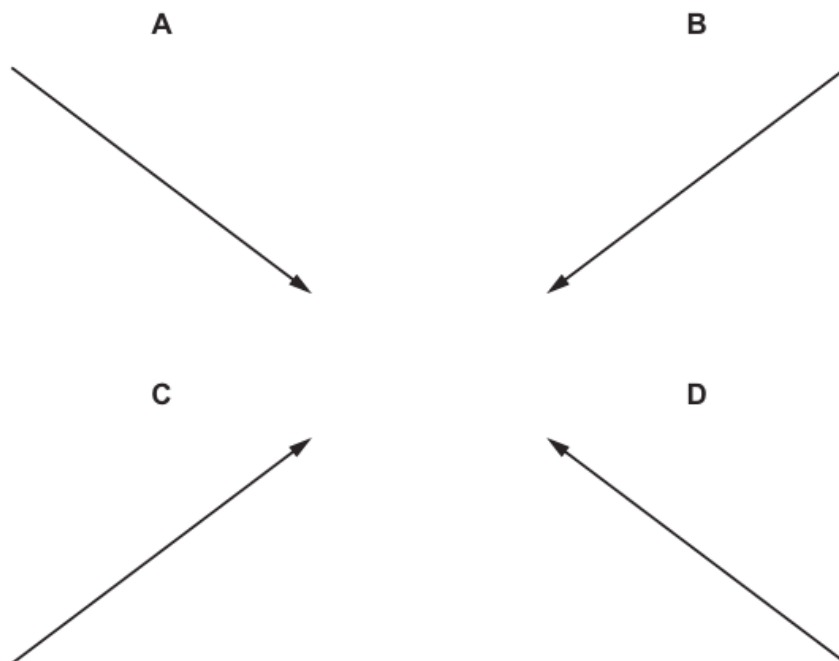
- A C
- B K
- C $^\circ\text{C}$
- D $^\circ\text{K}$

214. 9702/12/M/J/22/Q4

An object is moving with an initial velocity of 4.0 m s^{-1} to the right. The velocity of the object changes so that its final velocity is 3.0 m s^{-1} downwards, as shown.



Which arrow represents the change in velocity of the object?



215. 9702/13/M/J/22/Q1

Which pair of quantities are physical quantities?

- A charge and ampere
- B efficiency and kilogram
- C pascal and strain
- D period and potential difference

216. 9702/13/M/J/22/Q2

Which list of unit prefixes decreases in magnitude from left to right?

- A** centi, deci, milli
- B** deci, milli, centi
- C** pico, kilo, milli
- D** kilo, milli, pico

217. 9702/13/M/J/22/Q3

The drag coefficient C_d is a number with no units. It is used to compare the drag on different cars at different speeds. C_d is given by the equation

$$C_d = \frac{2F}{v^n \rho A}$$

where F is the drag force on the car, ρ is the density of the air, A is the cross-sectional area of the car and v is the speed of the car.

What is the value of n ?

- A** 1
- B** 2
- C** 3
- D** 4

218. 9702/13/M/J/22/Q5

Forces of magnitudes 2 N, 4 N and 7 N combine to produce a resultant force.

The magnitudes of the three forces are fixed, but the forces may act in any direction in the same plane.

What is **not** a possible magnitude of the resultant force?

- A** 0 N
- B** 5 N
- C** 8 N
- D** 13 N

219. 9702/11/O/N/22/Q1

What is needed to accurately represent all physical quantities?

- A** a base unit and a number
- B** a unit and a number expressed in standard form (scientific notation)
- C** a unit and a numerical magnitude
- D** an SI unit and a numerical magnitude

220. 9702/11/O/N/22/Q2

A voltmeter connected across a resistor in a circuit reads 3.6 V.

What could be the current in the resistor and the resistance of the resistor?

	current	resistance
A	150 mA	0.24 k Ω
B	15 mA	2.4 k Ω
C	1.5 mA	0.24 M Ω
D	15 μ A	240 k Ω

221. 9702/11/O/N/22/Q4

Which quantity is a vector?

- A momentum
- B speed
- C temperature
- D Young modulus

222. 9702/12/O/N/22/Q1

Which quantity is a physical quantity?

- A flavour
- B kelvin
- C minute
- D potential difference

223. 9702/12/O/N/22/Q2

What is a power of 3.7 MW when expressed in kilowatts?

- A $3.7 \times 10^{-3} \text{ kW}$
- B $3.7 \times 10^{-3} \text{ KW}$
- C $3.7 \times 10^3 \text{ kW}$
- D $3.7 \times 10^3 \text{ KW}$

224. 9702/12/O/N/22/Q4

What is the difference between a scalar quantity and a vector quantity?

- A A scalar quantity has direction but a vector quantity does not.
- B A scalar quantity has magnitude but a vector quantity does not.
- C A vector quantity has direction but a scalar quantity does not.
- D A vector quantity has magnitude but a scalar quantity does not.

225. 9702/13/O/N/22/Q1

A train of mass 600 000 kg moves with a speed of 100 km h^{-1} .

What is the order of magnitude of the kinetic energy of the train?

- A 10^6 J B 10^8 J C 10^{10} J D 10^{12} J

226. 9702/13/O/N/22/O2

What are the SI base units of electromotive force (e.m.f.)?

- A $\text{kg m}^2 \text{ s}^{-1} \text{ A}^{-1}$
- B $\text{kg m}^2 \text{ s}^{-3} \text{ A}^{-1}$
- C $\text{kg m}^2 \text{ s}^{-1} \text{ A}$
- D $\text{kg m s}^{-3} \text{ A}^{-1}$

227. 9702/13/O/N/22/Q4

Which quantity is a vector quantity?

- A density
- B mass
- C volume
- D weight

228. 9702/12/F/M/23/Q1

What represents a physical quantity?

- A 3.0
- B kilogram
- C 7.0 N
- D 40%

229. 9702/12/F/M/23/O2

The relationship between the variables D and T is given by the equation

$$\frac{1}{T} = \frac{b}{\sqrt{D}} + c$$

where b and c are constants.

The unit of D is m^2 and the unit of T is s.

What are the units of b and c ?

	unit of b	unit of c
A	ms	s
B	ms^{-1}	s^{-1}
C	m^{-1}s	s
D	$\text{m}^{-1}\text{s}^{-1}$	s^{-1}

230. 9702/11/M/J/23/Q1

Which unit is **not** an SI base unit?

- A A
- B kg
- C C
- D s

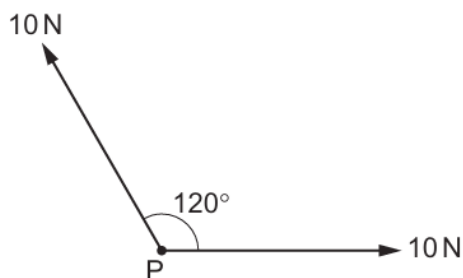
231. 9702/11/M/J/23/Q2

What is the best estimate of the number of atoms in a piece of metal of volume 50 cm^3 ?

- A 5×10^{15}
- B 5×10^{25}
- C 5×10^{29}
- D 5×10^{31}

232. 9702/11/M/J/23/Q4

Two forces, each of 10 N, act at a point P, as shown. The angle between the directions of the forces is 120° .



What is the magnitude of the resultant force?

- A** 5 N **B** 10 N **C** 17 N **D** 20 N

233. 9702/12/M/J/23/Q1

A stone sinks in water.

What is a possible value for the density of the stone?

- A** $8 \times 10^2 \text{ kg m}^{-3}$
B $2 \times 10^3 \text{ kg m}^{-3}$
C $8 \times 10^3 \text{ N m}^{-3}$
D $2 \times 10^4 \text{ N m}^{-3}$

234. 9702/12/M/J/23/Q2

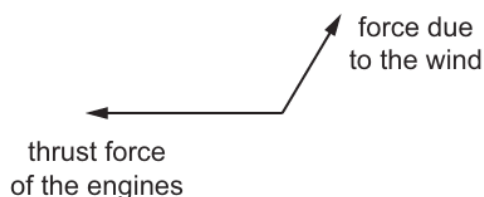
Gm, Tm, μm and pm are all units of length.

Which unit is the largest and which unit is the smallest?

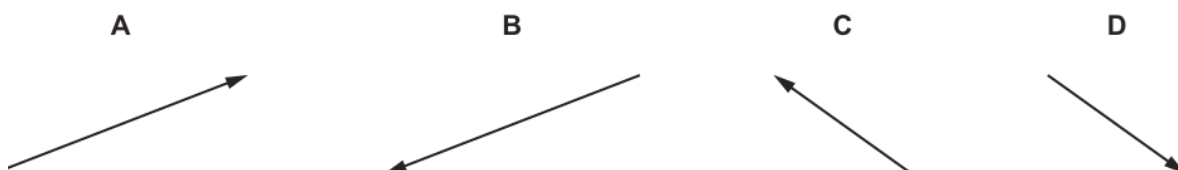
	largest unit	smallest unit
A	Gm	μm
B	Gm	pm
C	Tm	μm
D	Tm	pm

235. 9702/12/M/J/23/Q4

An aircraft travels along a horizontal path. Two of the forces that act horizontally on the aircraft are the thrust force of the engines and the force due to the wind. The vector diagram for these forces is shown.



Which vector represents the resultant horizontal force acting on the aircraft due to these two forces?



236. 9702/13/M/J/23/Q1

What **must** be included in a record of a physical quantity?

- A an integer value for the quantity
- B an SI unit
- C a numerical value for the quantity
- D a unit expressed in base units

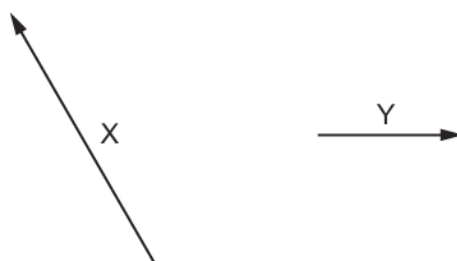
237. 9702/13/M/J/23/Q2

What is the ohm expressed in SI base units?

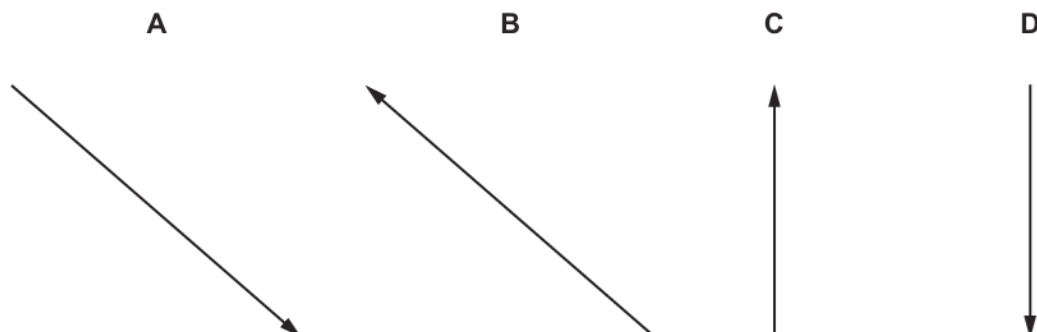
- A $\text{kg m}^2 \text{s}^{-3} \text{A}^{-2}$
- B $\text{kg}^{-1} \text{m}^{-2} \text{s}^3 \text{A}^2$
- C $\text{J C}^{-1} \text{A}^{-1}$
- D W A^{-2}

238. 9702/13/M/J/23/Q4

The diagram shows two vectors, X and Y, drawn to scale.



If $X = Y - Z$, which diagram represents the vector Z?



239. 9702/11/O/N/23/Q1

What is a reasonable estimate of the cross-sectional area of the wire in a paper clip?

- A** $1 \times 10^{-3} \text{ m}^2$ **B** $8 \times 10^{-5} \text{ m}^2$ **C** $8 \times 10^{-7} \text{ m}^2$ **D** $1 \times 10^{-9} \text{ m}^2$

240. 9702/11/O/N/23/Q2

Which quantity is **not** an SI base quantity?

- A** charge
B mass
C temperature
D time

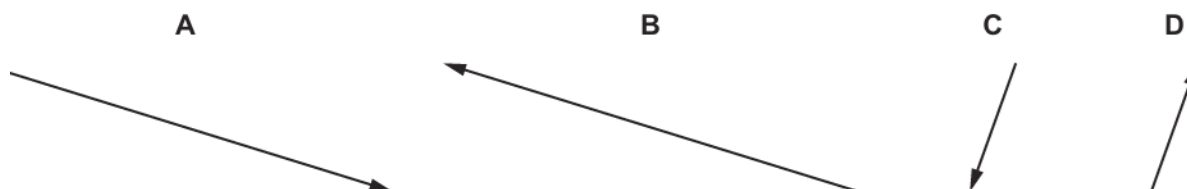
241. 9702/11/O/N/23/Q4

The diagram shows two coplanar forces, P and Q , drawn to scale.



Force R is given by $R = Q - P$.

Which diagram represents R ?



242. 9702/12/O/N/23/Q1

A student estimates the maximum speed of some different moving objects.

Which maximum speed is **not** a reasonable estimate?

- A** container ship: 10 m s^{-1}
B Olympic sprinter: 0.1 km s^{-1}
C racing car: 9000 cm s^{-1}
D snail: 0.01 km h^{-1}

243. 9702/12/O/N/23/Q2

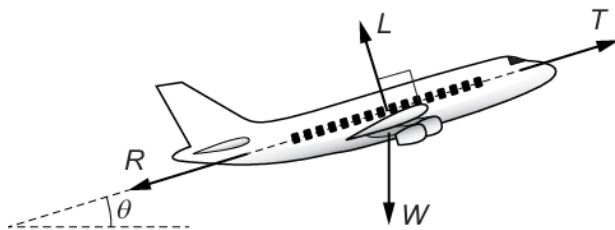
Which quantity is an SI base quantity?

- A** force
B newton
C second
D time

244. 9702/12/O/N/23/Q4

An aeroplane is moving at a constant speed in a straight line at an angle θ to the horizontal.

Four forces act on the aeroplane: thrust force T , weight W , lift force L and resistive force R .



Which two equations must be correct?

- A $L = W \cos \theta$ and $T = R + W \sin \theta$
- B $L = W \sin \theta$ and $T = R + W \cos \theta$
- C $L = W \cos \theta$ and $T = R - W \sin \theta$
- D $L = W \sin \theta$ and $T = R - W \cos \theta$

245. 9702/13/O/N/23/Q1

What is the best estimate of the wavelength of green light?

- A 260 nm B 540 nm C 780 nm D 920 nm

246. 9702/13/O/N/23/Q2

In an electric circuit, an ammeter reads $2 \mu\text{A}$.

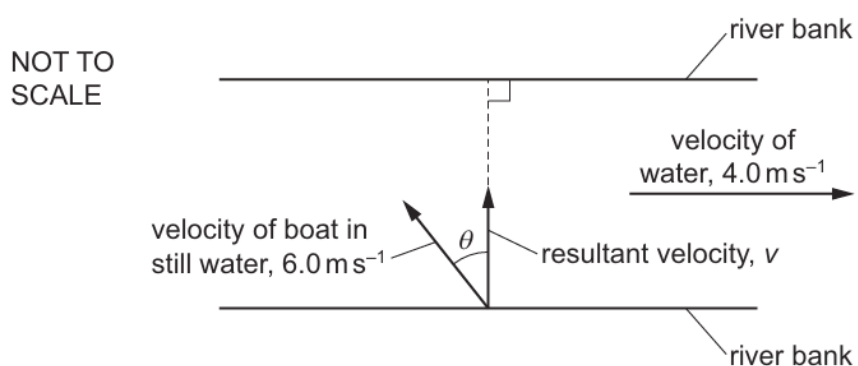
In a second circuit, the ammeter reads 1 mA.

How many times larger is the current in the second circuit compared with the current in the first circuit?

- A 500 B 5000 C 500 000 D 5 000 000

247. 9702/13/O/N/23/Q4

A boat is crossing a river in which the water is moving at a speed of 4.0 m s^{-1} from left to right.



In still water, the speed of the boat is 6.0 m s^{-1} . The boat is directed at an angle θ to a line perpendicular to the river banks. The resultant velocity v of the boat is in a direction perpendicular to the river banks.

What are the values of θ and v ?

	$\theta / ^\circ$	$v / \text{m s}^{-1}$
A	42	4.5
B	42	7.2
C	48	4.5
D	48	7.2

ANSWERS

1. B	48. D	95. C	142. B
2. B	49. C	96. B	143. B
3. C	50. B	97. C	144. A
4. B	51. A	98. D	145. C
5. B	52. B	99. A	146. C
6. B	53. A	100. B	147. A
7. C	54. C	101. C	148. C
8. C	55. C	102. A	149. B
9. A	56. C	103. A	150. B
10. B	57. C	104. C	151. D
11. A	58. B	105. A	152. A
12. A	59. A	106. C	153. C
13. C	60. D	107. B	154. C
14. C	61. B	108. A	155. D
15. D	62. C	109. C	156. C
16. C	63. C	110. D	157. A
17. C	64. D	111. C	158. A
18. B	65. C	112. C	159. A
19. B	66. D	113. C	160. B
20. C	67. C	114. D	161. C
21. D	68. A	115. D	162. D
22. B	69. D	116. D	163. C
23. D	70. B	117. D	164. C
24. D	71. A	118. C	165. D
25. D	72. B	119. B	166. D
26. B	73. C	120. C	167. C
27. A	74. B	121. D	168. B
28. C	75. D	122. A	169. B
29. D	76. C	123. C	170. D
30. A	77. D	124. D	171. C
31. C	78. C	125. C	172. C
32. A	79. C	126. C	173. C
33. B	80. D	127. A	174. A
34. C	81. B	128. A	175. A
35. D	82. B	129. C	176. C
36. C	83. B	130. C	177. B
37. C	84. A	131. C	178. A
38. B	85. B	132. B	179. C
39. A	86. C	133. B	180. A
40. D	87. B	134. B	181. B
41. C	88. D	135. A	182. A
42. C	89. B	136. B	183. B
43. B	90. D	137. B	184. A
44. A	91. A	138. D	185. D
45. B	92. A	139. D	186. C
46. D	93. B	140. D	
47. A	94. A	141. B	

187. C	231. B
188. A	232. B
189. D	233. B
190. C	234. D
191. D	235. C
192. D	236. C
193. B	237. A
194. D	238. A
195. C	239. C
196. C	240. A
197. A	241. C
198. C	242. B
199. A	243. D
200. D	244. A
201. B	245. B
202. B	246. A
203. A	247. A
204. D	
205. C	
206. B	
207. B	
208. B	
209. D	
210. B	
211. C	
212. C	
213. B	
214. B	
215. D	
216. D	
217. B	
218. A	
219. C	
220. D	
221. A	
222. D	
223. C	
224. C	
225. B	
226. B	
227. D	
228. C	
229. B	
230. C	